MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, SCHOTTKY TTL, ARITHMETIC LOGIC UNIT/FUNCTION GENERATORS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Schottky TTL, arithmetic logic unit/function generators. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	4-bit arithmetic logic unit/function generator
02	Lookahead carry generator

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
J	D-3 (24-lead, 1/2" x 1-1/4"), dual-in-line package
K	F-6 (24-lead, 3/8" x 5/8"), flat package
Z	F-8 (24-lead, 1/4" x 3/8"), flat package

1.3 Absolute maximum ratings:

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	
Device type 01	990 mW dc
Device type 02	544 mW dc
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}): The thermal resistance shall be as specified on figure C-6 in Appendix C of MIL-M-38510.	
Junction temperature (T_J) 2/	+175°C

1/ Must withstand the added P_D due to short circuit condition (e.g., I_{OS}) at one output for 5 seconds duration.

2/ Maximum junction temperature shall not be exceeded except for the allowable short duration burn-in screening per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions:

Supply voltage range (V_{CC}) - - - - - - - - - -	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH}) - - -	2.0 V dc
Maximum low-level input voltage (V_{IL}) 2/ - -	0.8 V dc
Normalized fanout (each output) 3/	
Logical low level- - - - - - - - - - - - - -	10 maximum
Logical high level - - - - - - - - - - - - - -	20 maximum
Case operating temperature range (T_C)- - - -	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and pin designations. The terminal connections and pin designations shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Logic diagrams. The logic diagrams shall be as specified on figure 3.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

2/ V_{IL} = 0.7 V @ 125°C.
3/ A fanout of 20 normalized loads is provided to facilitate connection of unused inputs to used inputs.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V; V_{IL} = 0.8 V; 2/ V_{IN} = 2.0 V; I_{OH} = -1 mA	01, 02	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V; V_{IL} = 0.8 V; 2/ V_{IH} = 2.0 V; I_{OL} = 20 mA	01, 02	0.2	0.5	V 3/
Input diode clamp	V_{IC}	V_{CC} = 4.5 V; I_{IN} = -18 mA T_C = +25°C	01, 02		-1.2	V
Low-level input current at M input	I_{IL1}	V_{CC} = 5.5 V; V_{IN} = 0.5 V 4/	01	-1	-2	mA
Low-level input current at $\overline{B}$ input	I_{IL2}	V_{CC} = 5.5 V; V_{IN} = 0.5 V 4/	01	-3	-6	mA
Low-level input current at S_0 or S_3 input	I_{IL3}	V_{CC} = 5.5 V; V_{IN} = 0.5 V 4/	01	-4	-8	mA
Low-level input current at S_1 or S_2	I_{IL4}	V_{CC} = 5.5 V; V_{IN} = 0.5 V; 5/ $\overline{A}_0$, $\overline{A}_1$, $\overline{A}_2$, $\overline{A}_3$ = 5.5 V	01	-4	-8	mA
Low-level input current at C_n input	I_{IL5}	V_{CC} = 5.5 V; V_{IN} = 0.5 V 5/	01	-5	-10	mA
Low-level input current at $\overline{A}$ input	I_{IL6}	V_{CC} = 5.5 V; V_{IN} = 0.5 V 4/	01	-2	-4	mA
Low-level input current at $\overline{G}_1$ input	I_{IL1}	V_{CC} = 5.5 V; V_{IN} = 0.5 V, C_n = GND 4/	02	-8	-16	mA
Low-level input current at $\overline{P}_1$ input	I_{IL2}	V_{CC} = 5.5 V; V_{IN} = 0.5 V, C_n = GND 4/	02	-4	-8	mA
Low-level input current at $\overline{G}_0$ input	I_{IL3}	V_{CC} = 5.5 V; V_{IN} = 0.5 V, C_n = GND 4/	02	-7	-14	mA
Low-level input current at $\overline{P}_0$ input	I_{IL4}	V_{CC} = 5.5 V; V_{IN} = 0.5 V 4/	02	-4	-8	mA
Low-level input current at $\overline{G}_3$ input	I_{IL5}	V_{CC} = 5.5 V; V_{IN} = 0.5 V 4/	02	-4	-8	mA
Low-level input current at $\overline{P}_3$ input	I_{IL6}	V_{CC} = 5.5 V; V_{IN} = 0.5 V 4/	02	-2	-4	mA

See footnotes at end of table I.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
Low-level input current at C_n input	I_{IL7}	V_{CC} = 5.5 V; V_{IN} = 0.5 V 4/	02	-1	-2	mA
Low-level input current at $\overline{G}_2$ input	I_{IL8}	V_{CC} = 5.5 V; V_{IN} = 0.5 V, C_n = GND 4/	02	-7	-14	mA
Low-level input current at $\overline{P}_2$ input	I_{IL9}	V_{CC} = 5.5 V; V_{IN} = 0.5 V 4/	02	-3	-6	mA
High-level input current at M input	I_{IH1}	V_{CC} = 5.5 V; V_{IN} = 2.7 V 5/	01		50	µA
High-level input current at $\overline{A}$ or $\overline{B}$ input	I_{IH2}	V_{CC} = 5.5 V; V_{IN} = 2.7 V 5/	01		150	µA
High-level input current at S_0 or S_3 input	I_{IH3}	V_{CC} = 5.5 V; V_{IN} = 2.7 V 5/	01		200	µA
High-level input current at S_1 or S_2 input	I_{IH4}	V_{CC} = 5.5 V; V_{IN} = 2.7 V; 5/ $\overline{B}_0$, $\overline{B}_1$, $\overline{B}_2$, $\overline{B}_3$ = 5.5 V	01		200	µA
High-level input current at C_n input	I_{IH5}	V_{CC} = 5.5 V; V_{IN} = 2.7 V 4/	01		250	µA
High-level input current at M input	I_{IH6}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 5/	01		1	mA
High-level input current at $\overline{A}$ or $\overline{B}$ input	I_{IH7}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 5/	01		1	mA
High-level input current at S_0 or S_3 input	I_{IH8}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 5/	01		1	mA
High-level input current at S_1 or S_2 input	I_{IH9}	V_{CC} = 5.5 V; V_{IN} = 5.5 V; 5/ $\overline{B}_0$, $\overline{B}_1$, $\overline{B}_2$, $\overline{B}_3$ = 5.5 V	01		1	mA
High-level input current at C_n input	I_{IH10}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 4/	01		1	mA
High-level input current at $\overline{G}_1$ input	I_{IH1}	V_{CC} = 5.5 V; V_{IN} = 2.7 V; C_n = 5.5 V 5/	02		400	µA
High-level input current at $\overline{P}_1$ input	I_{IH2}	V_{CC} = 5.5 V; V_{IN} = 2.7 V C_n = 5.5 V 5/	02		200	µA

See footnotes at end of table I.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
High-level input current at $\overline{G}_0$ input	I_{IH3}	V_{CC} = 5.5 V; V_{IN} = 2.7 V; C_n = 5.5 V 5/	02		350	µA
High-level input current at $\overline{P}_0$ input	I_{IH4}	V_{CC} = 5.5 V; V_{IN} = 2.7 V 5/	02		200	µA
High-level input current at $\overline{G}_3$ input	I_{IH5}	V_{CC} = 5.5 V; V_{IN} = 2.7 V 5/	02		200	µA
High-level input current at $\overline{P}_3$ input	I_{IH6}	V_{CC} = 5.5 V; V_{IN} = 2.7 V 5/	02		100	µA
High-level input current at C_n input	I_{IH7}	V_{CC} = 5.5 V; V_{IN} = 2.7 V 5/	02		50	µA
High-level input current at $\overline{G}_2$ input	I_{IH8}	V_{CC} = 5.5 V; V_{IN} = 2.7 V; C_n = 5.5 V 5/	02		350	µA
High-level input current at $\overline{P}_2$ input	I_{IH9}	V_{CC} = 5.5 V; V_{IN} = 2.7 V 5/	02		150	µA
High-level input current at $\overline{G}_1$ input	I_{IH10}	V_{CC} = 5.5 V; V_{IN} = 5.5 V; C_n = 5.5 V 5/	02		1	mA
High-level input current at $\overline{P}_1$ input	I_{IH11}	V_{CC} = 5.5 V; V_{IN} = 5.5 V; C_n = 5.5 V 5/	02		1	mA
High-level input current at $\overline{G}_0$	I_{IH12}	V_{CC} = 5.5 V; V_{IN} = 5.5 V; C_n = 5.5 V 5/	02		1	mA
High-level input current at $\overline{P}_0$ input	I_{IH13}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 5/	02		1	mA
High-level input current at $\overline{G}_3$ input	I_{IH14}	V_{CC} = 5.5 V; V_{IN} = 5.5 V; C_n = GND 5/	02		1	mA
High-level input current at $\overline{P}_3$ input	I_{IH15}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 5/	02		1	mA
High-level input current at C_n input	I_{IH16}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 5/	02		1	mA
High-level input current at $\overline{G}_2$	I_{IH17}	V_{CC} = 5.5 V; V_{IN} = 5.5 V; C_n = 5.5 V 5/	02		1	mA

See footnotes at end of table I.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
High-level input current at $\overline{P}_2$ input	I_{IH18}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 5/	02		1	mA
Short-circuit output current at $\overline{G}$ output	I_{OS1}	V_{CC} = 5.5 V; S_3, $\overline{A}_3$, $\overline{B}_3$ = 5.5 V 5/ 6/	01	-40	-100	mA
Short-circuit output current at C_{n+4} output	I_{OS2}	V_{CC} = 5.5 V 5/ 6/	01	-40	-100	mA
Short-circuit output current at $\overline{P}$, $\overline{F}_0$ thru $\overline{F}_3$ outputs	I_{OS3}	V_{CC} = 5.5 V 4/ 6/	01	-40	-100	mA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V; V_{IL} = GND; V_{IN} = 5.5 V 6/	02	-40	-100	mA
Collector cutoff current	I_{CEX}	V_{CC} = 5.5 V; V_{OH} = 5.5 V; V_{IH} = 2.0 V; V_{IL} = GND	01		1	mA
		V_{CC} = 5.5 V; V_{OH} = 5.5 V; V_{IH} = 5.5 V; V_{IL} = GND	02		250	µA
Low-level supply current	I_{CCL}	V_{CC} = 5.5 V; V_{IL} = 0 V; V_{IH} = 5.5 V 7/	01		180	mA
	I_{CCL}	V_{CC} = 5.5 V; V_{IL} = 0 V; V_{IH} = 5.5 V	02		99	mA
High-level supply current	I_{CCH}	V_{CC} = 5.5 V; V_{IL} = 0 V; V_{IH} = 5.5 V 8/	01		180	mA
	I_{CCH}	V_{CC} = 5.5 V; V_{IL} = 0 V; V_{IH} = 5.5 V	02		65	mA
Propagation delay times, high-to-low level output: 9/						
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{F}$ $\overline{sum}$ mode	t_{PHL1}	V_{CC} = 5.0 V See figure 5 $\overline{sum}$ mode test table	01	2	23	ns
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{F}$ $\overline{diff}$ mode	t_{PHL2}	V_{CC} = 5.0 V See figure 5 $\overline{diff}$ mode test table	01	2	28	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{P}$ $\overline{sum}$ mode	t_{PHL3}	V_{CC} = 5.0 V See figure 5 $\overline{sum}$ mode test table	01	2	16	ns

See footnotes at end of table I.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
Propagation delay times, high-to-low level output: 9/						
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{P}$ $\overline{diff}$ mode	t_{PHL4}	V_{CC} = 5.0 V See figure 5 $\overline{diff}$ mode test table	01	2	20	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{G}$ $\overline{sum}$ mode	t_{PHL5}	V_{CC} = 5.0 V See figure 5 $\overline{sum}$ mode test table	01	2	16	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{G}$ $\overline{diff}$ mode	t_{PHL6}	V_{CC} = 5.0 V See figure 5 $\overline{diff}$ mode test table	01	2	20	ns
C_n to any $\overline{F}_0$ $\overline{sum}$ mode	t_{PHL7}	V_{CC} = 5.0 V See figure 5 $\overline{sum}$ mode test table	01	2	16	ns
$\overline{A}_i$ or $\overline{B}_i$ to A = B $\overline{diff}$ mode	t_{PHL8}	V_{CC} = 5.0 V See figure 5 $\overline{diff}$ mode test table	01	2	36	ns
C_n to C_{n+4} $\overline{sum}$ mode	t_{PHL9}	V_{CC} = 5.0 V See figure 5 $\overline{sum}$ mode test table	01	2	14.5	ns
C_n to C_{n+4} $\overline{diff}$ mode	t_{PHL10}	V_{CC} = 5.0 V See figure 5 $\overline{diff}$ mode test table	01	2	14.5	ns
$\overline{A}_i$ or $\overline{B}_i$ to C_{n+4} $\overline{sum}$ mode	t_{PHL11}	V_{CC} = 5.0 V See figure 5 $\overline{sum}$ mode test table	01	2	25	ns
$\overline{A}_i$ or $\overline{B}_i$ to C_{n+4} $\overline{diff}$ mode	t_{PHL12}	V_{CC} = 5.0 V See figure 5 $\overline{diff}$ mode test table	01	2	29	ns
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{F}$ logic mode	t_{PHL13}	V_{CC} = 5.0 V See figure 5 logic mode test table	01	2	28	ns
Propagation delay times, low-to-high level output: 9/						
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{F}$ $\overline{sum}$ mode	t_{PLH1}	V_{CC} = 5.0 V See figure 5 $\overline{sum}$ mode test table	01	2	23	ns
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{F}$ $\overline{diff}$ mode	t_{PLH2}	V_{CC} = 5.0 V See figure 5 $\overline{diff}$ mode test table	01	2	26	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{P}$ $\overline{sum}$ mode	t_{PLH3}	V_{CC} = 5.0 V See figure 5 $\overline{sum}$ mode test table	01	2	16	ns

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
Propagation delay times, low-to-high level output: 9/						
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{F}$ $\overline{diff}$ mode	t_{PLH4}	V_{CC} = 5.0 V See figure 5 $\overline{diff}$ mode test table	01	2	20	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{G}$ $\overline{sum}$ mode	t_{PLH5}	V_{CC} = 5.0 V See figure 5 $\overline{sum}$ mode test table	01	2	16	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{G}$ $\overline{diff}$ mode	t_{PLH6}	V_{CC} = 5.0 V See figure 5 $\overline{diff}$ mode test table	01	2	20	ns
C_n to any $\overline{F}$ $\overline{sum}$ mode	t_{PLH7}	V_{CC} = 5.0 V See figure 5 $\overline{sum}$ mode test table	01	2	16	ns
A_i or B_i to A = B $\overline{diff}$ mode	t_{PLH8}	V_{CC} = 5.0 V See figure 5 $\overline{diff}$ mode test table	01	2	29	ns
C_n to C_{n+4} $\overline{sum}$ mode	t_{PLH9}	V_{CC} = 5.0 V See figure 5 $\overline{sum}$ mode test table	01	2	14.5	ns
C_n to C_{n+4} $\overline{diff}$ mode	t_{PLH10}	V_{CC} = 5.0 V See figure 5 $\overline{diff}$ mode test table	01	2	14.5	ns
$\overline{A}_i$ or $\overline{B}_i$ to C_{n+4} $\overline{sum}$ mode	t_{PLH11}	V_{CC} = 5.0 V See figure 5 $\overline{sum}$ mode test table	01	2	25	ns
$\overline{A}_i$ or $\overline{B}_i$ to C_{n+4} $\overline{diff}$ mode	t_{PLH12}	V_{CC} = 5.0 V See figure 5 $\overline{diff}$ mode test table	01	2	29	ns
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{F}$ logic mode	t_{PLH13}	V_{CC} = 5.0 V See figure 5 logic mode test table	01	2	26	ns
Propagation delay times, high-to-low level output: 10/						
C_n to C_{n+X}, C_{n+Y}, C_{n+Z}	t_{PHL1}	V_{CC} = 5.0 V; $\overline{P}_0$, $\overline{P}_1$, $\overline{P}_2$ = GND; $\overline{G}_i$ = 2.7 V	02	2	14.5	ns
$\overline{P}_0$, $\overline{P}_1$, or $\overline{P}_2$, to C_{n+X}, C_{n+Y}, C_{n+Z}	t_{PHL2}	V_{CC} = 5.0 V; $\overline{P}_0$, $\overline{P}_1$, $\overline{P}_2$ = GND; C_n, $\overline{G}_i$ = 2.7 V	02	2	11	ns
$\overline{G}_0$, $\overline{G}_1$, or $\overline{G}_2$ to C_{n+Y}, C_{n+X}, C_{n+Z}	t_{PHL3}	V_{CC} = 5.0 V; C_n, $\overline{P}_0$, $\overline{P}_1$, $\overline{P}_2$ = GND, $\overline{G}_i$ = 2.7 V 11/	02	2	11	ns

See footnotes at end of table I.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
Propagation delay times, high-to-low level output: 10/						
$\overline{P}_1$, $\overline{P}_2$, or $\overline{P}_3$ to $\overline{G}$	t_{PHL4}	V_{CC} = 5.0 V; $\overline{P}_i$ = GND; 11/ C_n, $\overline{G}_1$, $\overline{G}_2$, $\overline{G}_3$ = 2.7 V	02	2	14.5	ns
$\overline{G}_0$, $\overline{G}_1$, $\overline{G}_2$, or $\overline{G}_3$ to $\overline{G}$	t_{PHL5}	V_{CC} = 5.0 V; $\overline{G}_i$ = 2.7 V; 11/ $\overline{P}_1$, $\overline{P}_2$, $\overline{P}_3$ = GND	02	2	14.5	ns
$\overline{P}_0$, $\overline{P}_1$, $\overline{P}_2$, or $\overline{P}_3$ to $\overline{P}$	t_{PHL6}	$\overline{P}_i$ = GND 11/, V_{CC} = 5.0 V; C_n, $\overline{G}_i$ = 2.7 V	02	2	14	ns
Propagation delay times, low-to-high level output: 10/						
C_n to C_{n+X}, C_{n+Y}, C_{n+Z}	t_{PLH1}	V_{CC} = 5.0 V; $\overline{P}_0$, $\overline{P}_1$, $\overline{P}_2$ = GND; $\overline{G}_i$ = 2.7 V	02	2	14	ns
$\overline{P}_0$, $\overline{P}_1$, or $\overline{P}_2$ to C_{n+X}, C_{n+Y}, C_{n+Z}	t_{PLH2}	V_{CC} = 5.0 V; C_n, $\overline{G}_i$ = 2.7 V; $\overline{P}_0$, $\overline{P}_1$, $\overline{P}_2$ = GND 11/	02	2	11	ns
$\overline{G}_0$, $\overline{G}_1$, or $\overline{G}_2$ to C_{n+X}, C_{n+Y}, C_{n+Z}	t_{PLH3}	V_{CC} = 5.0 V; C_n, $\overline{P}_0$, $\overline{P}_1$, $\overline{P}_2$ = GND; $\overline{G}_i$ = 2.7 V 11/	02	2	11	ns
$\overline{P}_1$, $\overline{P}_2$, or $\overline{P}_3$, to $\overline{G}$	t_{PLH4}	V_{CC} = 5.0 V; C_n, $\overline{G}_1$, $\overline{G}_2$, $\overline{G}_3$ = 2.7 V; P_i = GND 11/	02	2	11.5	ns
$\overline{G}_0$, $\overline{G}_1$, $\overline{G}_2$, or $\overline{G}_3$ to $\overline{G}$	t_{PLH5}	V_{CC} = 5.0 V; $\overline{P}_1$, $\overline{P}_2$, $\overline{P}_3$ = GND; $\overline{G}_i$ = 2.7 V 11/	02	2	11.5	ns
$\overline{P}_0$, $\overline{P}_1$, $\overline{P}_2$, or $\overline{P}_3$ to $\overline{P}$	t_{PLH6}	V_{CC} = 5.0 V; $\overline{P}_i$ = GND 11/ C_n, $\overline{G}_i$ = 2.7 V	02	2	10	ns

1/ Condition of inputs specified in table III.

2/ V_{IL} = 0.7 V dc at 125°C.

3/ $V_{OL(max)}$ = 0.45 V dc at 125°C.

4/ All unspecified inputs at 5.5 V.

5/ All unspecified inputs at 0 V.

6/ Not more than one output should be shorted at a time.

7/ S_i, M, $\overline{A}_i$ at 4.5 V; all other inputs grounded; outputs open.

8/ S_i, M at 4.5 V; all other inputs grounded; outputs open.

9/ In $\overline{A}_i$ and $\overline{B}_i$, i = 0, 1, 2, or 3.

10/ In $\overline{G}_i$ and $\overline{P}_i$, i = 0, 1, 2, or 3.

11/ If not under test.

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3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are specified in table I, and apply over the full recommended case operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 11 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*, 2, 3, 7, 9, 10, 11	1*, 2, 3, 7, 9
Group A test requirements (method 5005)	1, 2, 3, 7, 8, 9, 10, 11	1, 2, 3, 7, 9, 10, 11
Group B test requirements (method 5005) subgroup 5	1, 2, 3, 7, 8, 9, 10, 11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1, 2, 3
Additional electrical subgroups for group C periodic inspections	N/A	None
Group D end-point electrical parameters (method 5005)	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1 (see 4.3c.).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.3 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

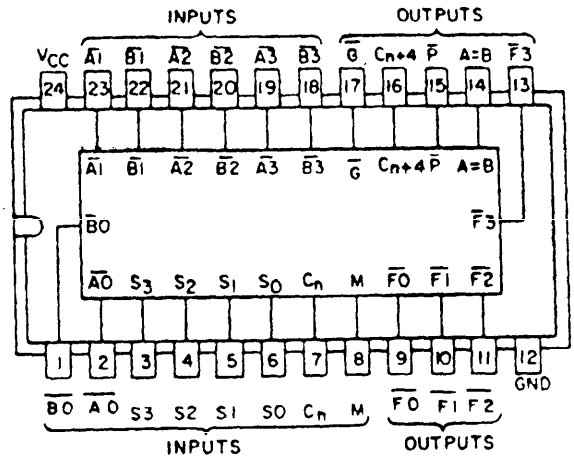
(1) Test condition D, using the circuit shown on figure 4, or equivalent.
(2) T_A = +125°C, minimum.

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Pin designations

Designation	Pin nos.	Function
$\overline{A}3, \overline{A}2, \overline{A}1, \overline{A}0$	19, 21, 23, 2	Word A inputs
$\overline{B}3, \overline{B}2, \overline{B}1, \overline{B}0$	18, 20, 22, 1	Word B inputs
S3, S2, S1, S0	3, 4, 5, 6	Function-select inputs
C_n	7	Inv. carry input
M	8	Mode control input
$\overline{F}3, \overline{F}2, \overline{F}1, \overline{F}0$	13, 11, 10, 9	Function outputs
A = B	14	Comparator output
$\overline{P}$	15	Carry propagate output
C_{n+4}	16	Inv. carry output
$\overline{G}$	17	Carry generate output
V_{CC}	24	Supply voltage
GND	12	Ground

Cases J, K, and Z

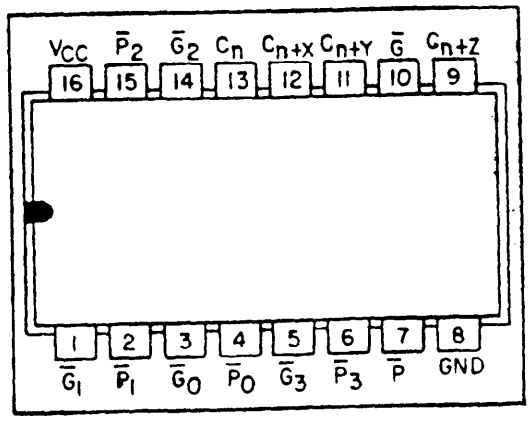


Device type 01

Pin designations

Designation	Pin nos.	Function
C_n	13	Carry input
$\overline{G}0, \overline{G}1, \overline{G}2, \overline{G}3$	3, 1, 14, 5	Carry generate input
$\overline{P}0, \overline{P}1, \overline{P}2, \overline{P}3$	4, 2, 15, 6	Carry propagate input
C_{n+x}, C_{n+y}, C_{n+z}	12, 11, 9	Carry output
$\overline{G}$	10	Carry generate output
$\overline{P}$	7	Carry propagate output

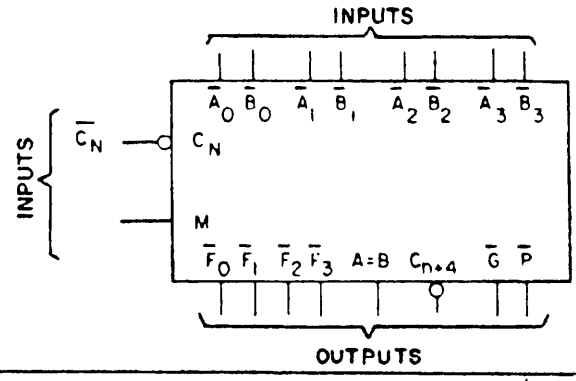
Cases E and F



Device type 02

FIGURE 1. Terminal connections and pin designations.

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	Selection $S_3S_2S_1S_0$	Active high data 3/		
		M = H Logic functions	M = L; Arithmetic operations	
			C_n = 0 $\overline{C_n}$ = 1 = H	C_n = 1 $\overline{C_n}$ = 0 = L
0	L L L L	F = $\overline{A}$	F = A	F = A plus 1
1	L L L H	F = $\overline{A + B}$	F = A + B	F = (A + B) plus 1
2	L L H L	F = $\overline{A}B$	F = A + $\overline{B}$	F = (A + $\overline{B}$) plus 1
3	L L H H	F = 0	F = minus 1 (2's compl)	F = zero
4	L H L L	F = $\overline{AB}$	F = A plus A$\overline{B}$	F = A plus A$\overline{B}$ plus 1
5	L H L H	F = $\overline{B}$	F = (A + B) plus A$\overline{B}$	F = (A + B) plus A$\overline{B}$ plus 1
2/ 6	L H H L	F = A $\oplus$ B	F = A minus B minus 1	F = A minus B
7	L H H H	F = A$\overline{B}$	F = A$\overline{B}$ minus 1	F = A$\overline{B}$
8	H L L L	F = $\overline{A}$ + B	F = A plus AB	F = A plus AB plus 1
9	H L L H	F = $\overline{A \oplus B}$	F = A plus B	F = A plus B plus 1
10	H L H L	F = B	F = (A + $\overline{B}$) plus AB	F = (A + $\overline{B}$) plus AB plus 1
11	H L H H	F = AB	F = AB minus 1	F = AB
12	H H L L	F = 1	F = A plus A 1/	F = A plus A plus 1
13	H H L H	F = A + $\overline{B}$	F = (A + B) plus A	F = (A + B) plus A plus 1
14	H H H L	F = A + B	F = (A + $\overline{B}$) plus A	F = (A + $\overline{B}$) plus A plus 1
15	H H H H	F = A	F = A minus 1	F = A

1/ Each bit is shifted to the next more significant position.

2/ This device (ALU) can be used as a comparator when placed in the subtract mode (i.e., $S_3S_2S_1S_0$ are at logical levels LHHL respectively) and the following expressions are valid:

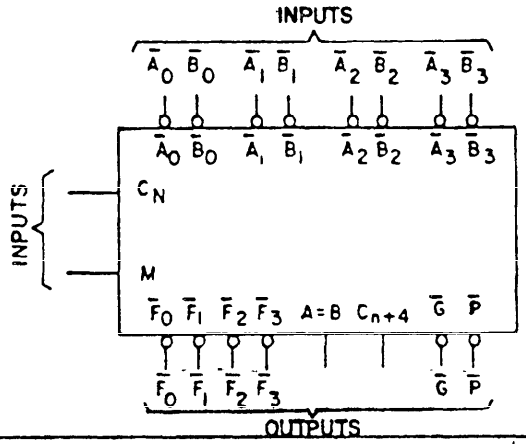
Active high data

When C_n is high and C_{n+4} is high, then A $\leq$ B
When C_n is low and C_{n+4} is high, then A < B
When C_n is high and C_{n+4} is low, then A > B
When C_n is low and C_{n+4} is low, then A $\geq$ B

3/ The table shown applies for positive logic. If negative logic is used, active high data becomes active low data.

FIGURE 2. Truth tables and logic equations for device type 01.

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	Selection $S_3S_2S_1S_0$	Active low data 5/		
		M = H Logic functions	M = L; Arithmetic operations	
			C_n = 0 = L	C_n = 1 = H
0	L L L L	F = $\overline{A}$	F = A minus 1	F = A
1	L L L H	F = $\overline{AB}$	F = AB minus 1	F = AB
2	L L H L	F = $\overline{A}$ + B	F = $A\overline{B}$ minus 1	F = $A\overline{B}$
3	L L H H	F = 1	F = minus 1 (2's compl)	F = zero
4	L H L L	F = $\overline{A + B}$	F = A plus (A + $\overline{B}$)	F = A plus (A + $\overline{B}$) plus 1
5	L H L H	F = $\overline{B}$	F = AB plus (A + $\overline{B}$)	F = AB plus (A + $\overline{B}$) plus 1
4/ 6	L H H L	F = $\overline{A \oplus B}$	F = A minus B minus 1	F = A minus B
7	L H H H	F = A + $\overline{B}$	F = A + $\overline{B}$	F = (A + $\overline{B}$) plus 1
8	H L L L	F = $\overline{A}$B	F = A plus (A + B)	F = A plus (A + B) plus 1
9	H L L H	F = A $\oplus$ B	F = A plus B	F = A plus B plus 1
10	H L H L	F = B	F = $A\overline{B}$ plus (A + B)	F = $A\overline{B}$ plus (A + B) plus 1
11	H L H H	F = A + B	F = A + B	F = (A + B) plus 1
12	H H L L	F = 0	F = A plus A 1/	F = A plus A plus 1
13	H H L H	F = $A\overline{B}$	F = AB plus A	F = AB plus A plus 1
14	H H H L	F = AB	F = $A\overline{B}$ plus A	F = $A\overline{B}$ plus A plus 1
15	H H H H	F = A	F = A	F = A plus 1

4/ This device (ALU) can be used as a comparator when placed in the subtract mode (i.e., $S_3S_2S_1S_0$ are at logical levels LHHL respectively) and the following expressions are valid:

Active low data

When C_n is low and C_{n+4} is low, then A $\leq$ B
When C_n is high and C_{n+4} is low, then A < B
When C_n is low and C_{n+4} is high, then A > B
When C_n is high and C_{n+4} is high, then A $\geq$ B

5/ The table shown applies for negative logic. If positive logic is used, active low data becomes active high data.

FIGURE 2. Truth tables and logic equations for device type 01 - Continued.

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Inputs									Outputs				
C_n	$\overline{G}_0$	$\overline{P}_0$	$\overline{G}_1$	$\overline{P}_1$	$\overline{G}_2$	$\overline{P}_2$	$\overline{G}_3$	$\overline{P}_3$	C_{n+x}	C_{n+y}	C_{n+z}	$\overline{G}$	$\overline{P}$
X	H	H							L				
L	H	X							L				
X	L	X							H				
H	X	L							H				
X	X	X	H	H						L			
X	H	H	H	X						L			
L	H	X	H	X						L			
X	X	X	L	X						H			
X	L	X	X	L						H			
H	X	L	X	L						H			
X	X	X	X	X	H	H					L		
X	X	X	H	H	H	X					L		
X	H	H	H	X	H	X					L		
L	H	X	H	X	H	X					L		
X	X	X	X	X	L	X					H		
X	X	X	L	X	X	L					H		
X	L	X	X	L	X	L					H		
H	X	L	X	L	X	L					H		
	X		X	X	X	X	H	H				H	
	X		X	X	H	H	H	X				H	
	X		H	H	H	X	H	X				H	
	H		H	X	H	X	H	X				H	
	X		X	X	X	X	L	X				L	
	X		X	X	L	X	X	L				L	
	X		L	X	X	L	X	L				L	
	L		X	L	X	L	X	L				L	
		H		X		X		X					H
		X		H		X		X					H
		X		X		H		X					H
		X		X		X		H					H
		L		L		L		L					L

H = High voltage level
L = Low voltage level
X = Don't care

FIGURE 2. Truth tables and logic equations for device type 02 - Continued.

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DEVICE TYPE 01

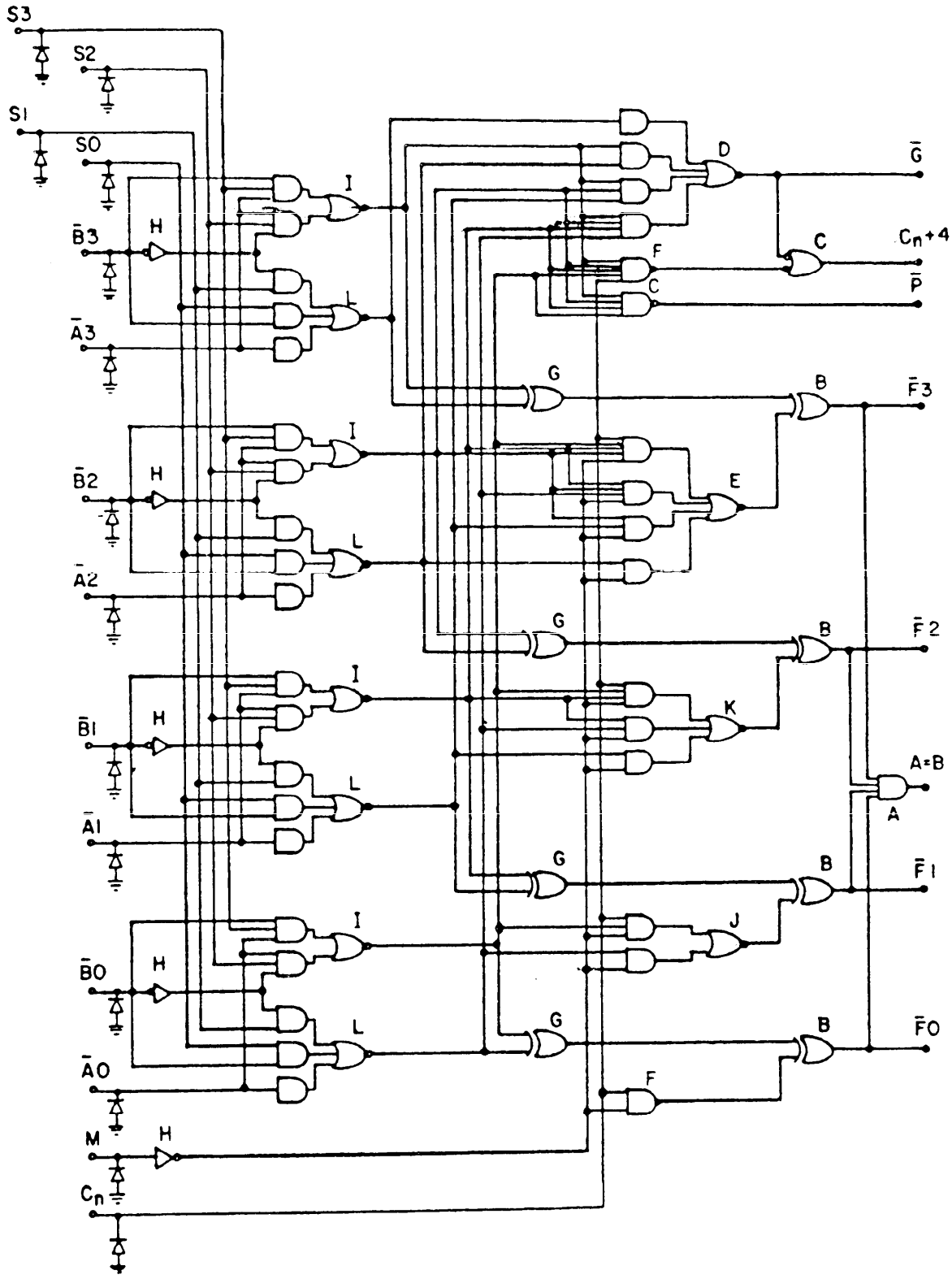


FIGURE 3. Logic diagrams.

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DEVICE TYPE 02

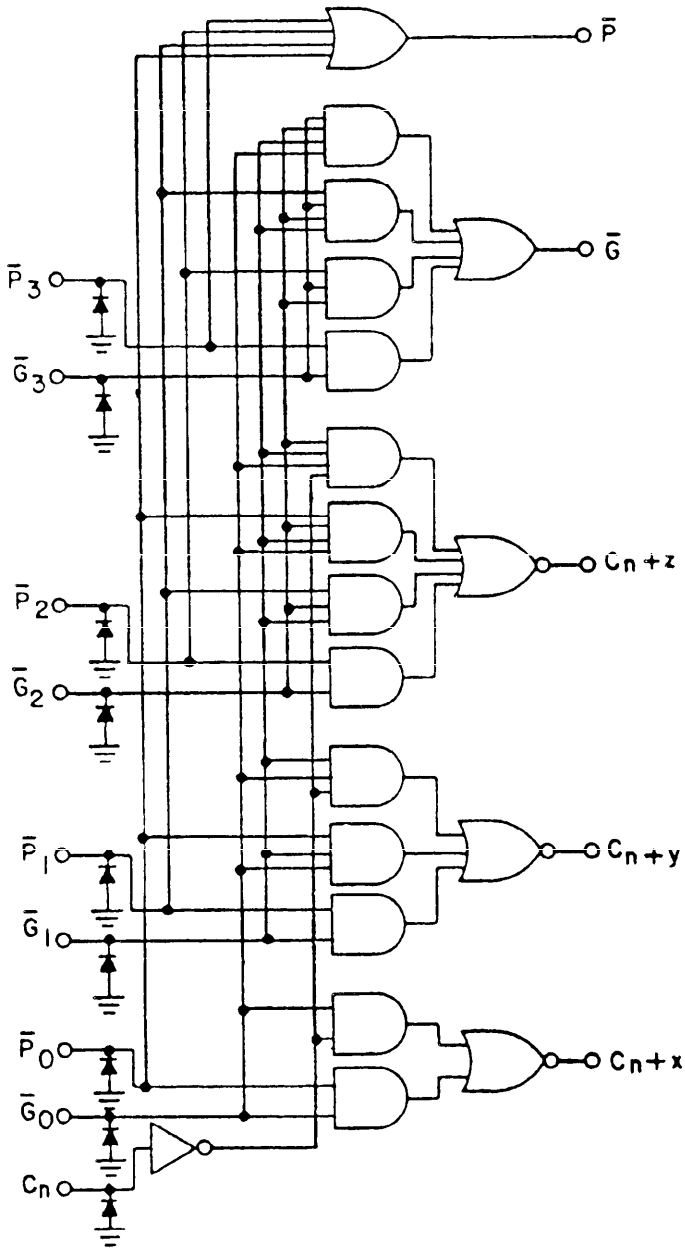
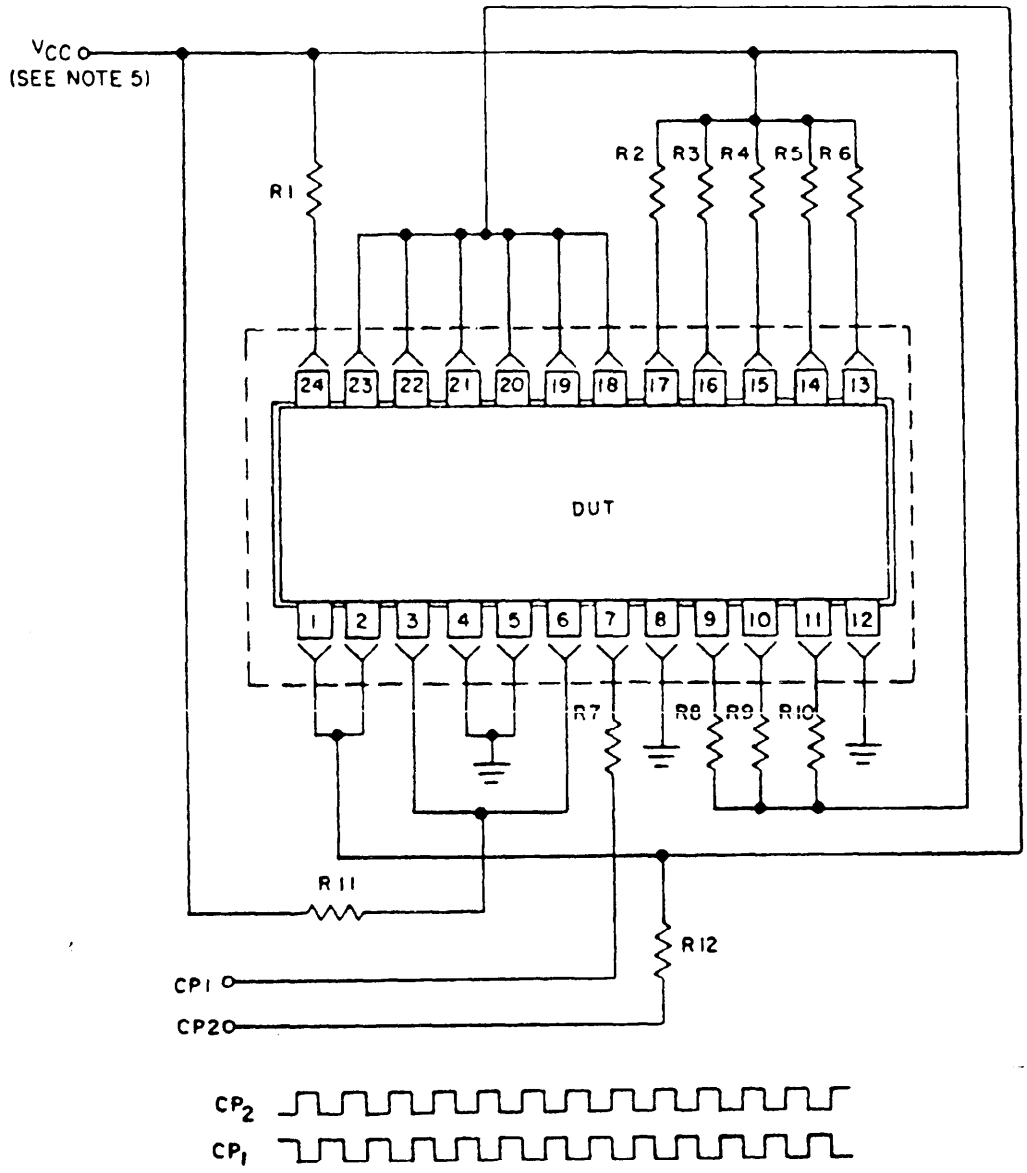


FIGURE 3. Logic diagrams - Continued.

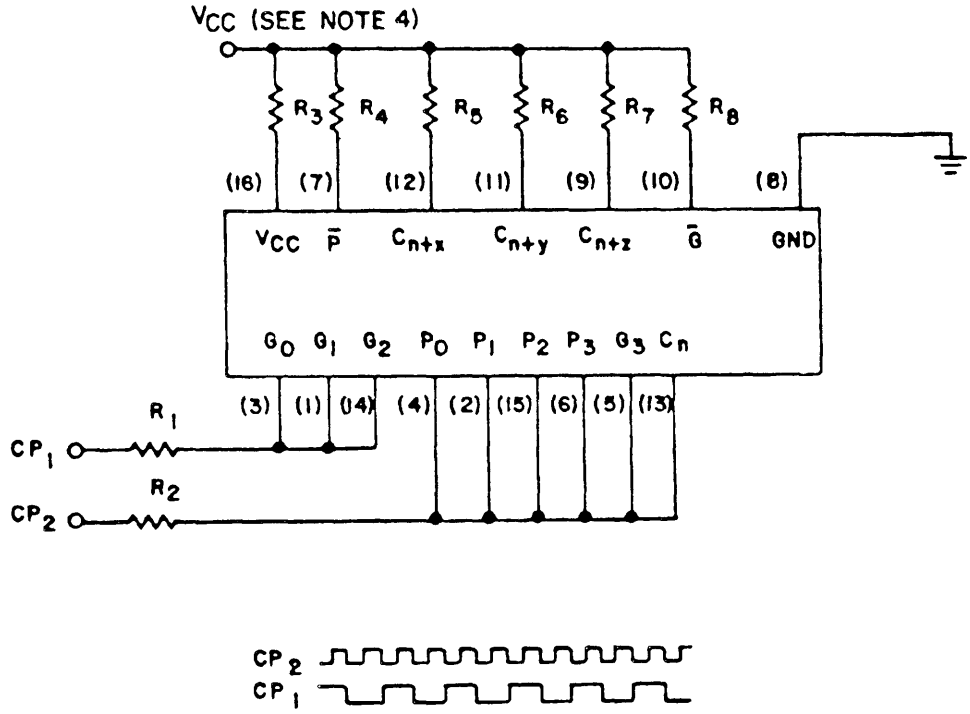
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NOTES:
1. R_1 = 5 ohms ±5%.
2. R_7 = 27 ohms ±5%.
3. R_2 thru R_6 = R_8 thru R_{11} = 180 ohms ±5%.
4. R_{12} = 82 ohms ±5%.
5. V_{CC} shall be high enough to insure 5 V minimum at device terminals.
6. CP_1 = 100 kHz ±10%, duty cycle = 50% ±10%, 3 V minimum at device terminals.
7. CP_2 = 100 kHz ±10%, duty cycle = 50% ±10%, 3 V minimum at device terminals.

FIGURE 4. Test circuit for burn-in and life test for device type 01.

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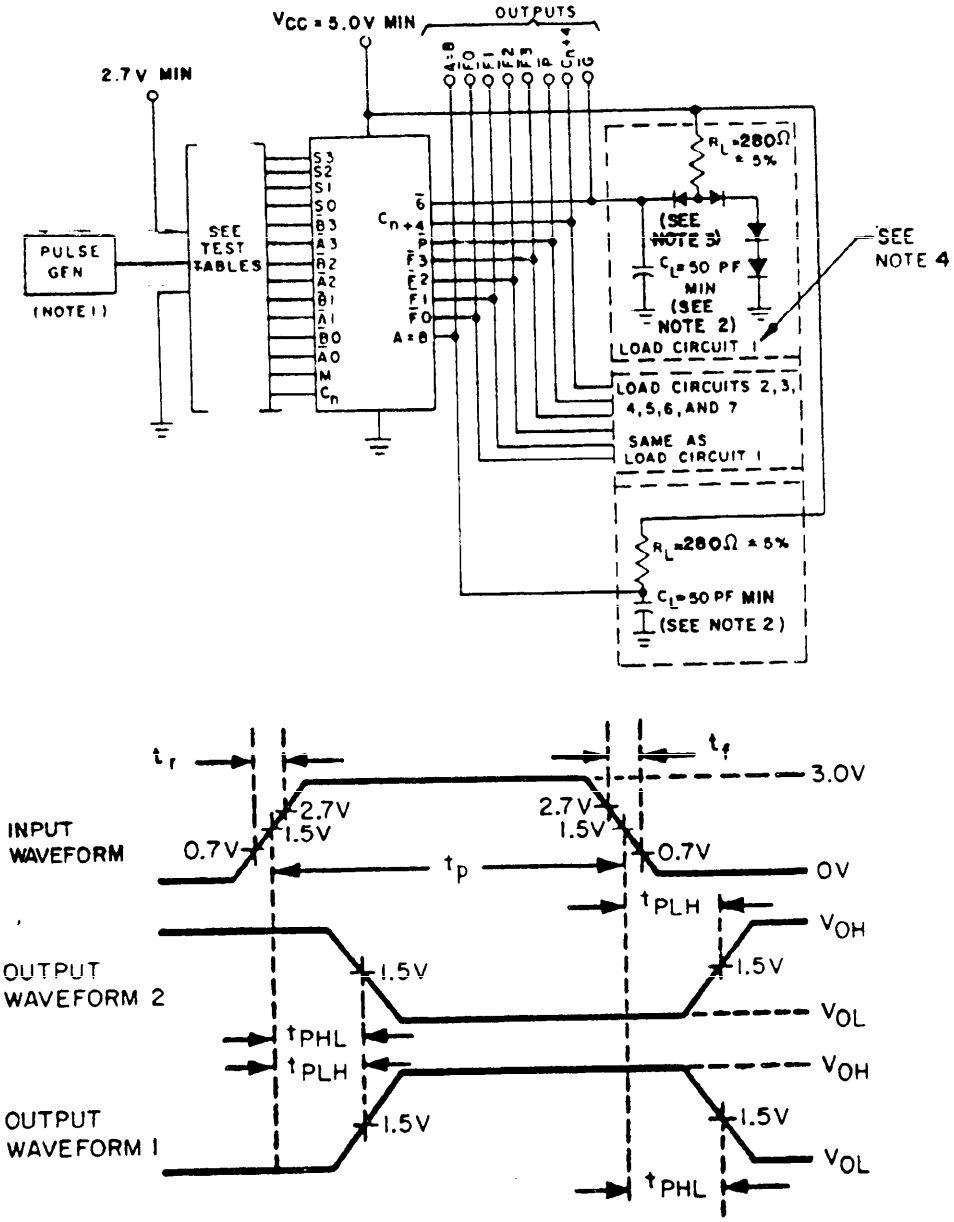


NOTES:
1. $R_1 = R_2 = 10\Omega \pm 5\%$.
2. $R_3 = 4.7\Omega \pm 5\%$.
3. R_4 thru $R_8 = 180\Omega \pm 5\%$.
4. V_{CC} shall be high enough to insure 5 V minimum at device terminals.
5. CP_1 = 100 kHz ±10%, duty cycle, 50% ±10%, 3 V minimum at device terminals.
6. CP_2 = 50 kHz ±10%, duty cycle, 50% ±10%, 3 V minimum at device terminals.

FIGURE 4. Test circuit for burn-in and life test for device type 02 - Continued.

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Qualification requirements have



NOTES:
1. The input pulse has the following characteristics: PRR $\leq$1 MHz, t_r = t_f $\leq$2.5 ns, $Z_{OUT} \approx 50\Omega$.
2. C_L includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. Load circuit is required on a given output only where table III indicates "OUT" on that output. Load circuits may otherwise be omitted.

FIGURE 5. Waveforms for propagation delay time and test circuit for device type 01.

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$\overline{SUM}$ MODE TEST TABLE

FUNCTION INPUTS: S0 = S3 = 2.7 V, S1 = S2 = M = 0 V

TEST	INPUT UNDER TEST	OTHER INPUT SAME BIT		OTHER DATA INPUTS		OUTPUT UNDER TEST	OUTPUT WAVEFORM
		APPLY 2.7 V	APPLY GND	APPLY 2.7 V	APPLY GND		
t_{PLH1} / t_{PHL1}	$\overline{A}_i$ 1/	$\overline{B}_i$	None	Remaining $\overline{A}$ and $\overline{B}$	C_n	Any $\overline{F}_i$	1
t_{PLH1} / t_{PHL1}	$\overline{B}_i$	$\overline{A}_i$	None	Remaining $\overline{A}$ and $\overline{B}$	C_n	Any $\overline{F}_i$	1
t_{PLH3} / t_{PHL3}	$\overline{A}_i$	$\overline{B}_i$	None	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	$\overline{P}$	1
t_{PLH3} / t_{PHL3}	$\overline{B}_i$	$\overline{A}_i$	None	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	$\overline{P}$	1
t_{PLH5} / t_{PHL5}	$\overline{A}_i$	None	$\overline{B}_i$	Remaining $\overline{B}$	Remaining $\overline{A}$, C_n	$\overline{G}$	1
t_{PLH5} / t_{PHL5}	$\overline{B}_i$	None	$\overline{A}_i$	Remaining $\overline{B}$	Remaining $\overline{A}$, C_n	$\overline{G}$	1
t_{PLH7} / t_{PHL7}	C_n	None	None	All $\overline{A}$	All $\overline{B}$	Any $\overline{F}_i$	1
t_{PLH9} / t_{PHL9}	C_n	None	None	All $\overline{A}$	All $\overline{B}$	C_{n+4}	1
t_{PLH11} / t_{PHL11}	$\overline{A}_i$	None	$\overline{B}_i$	Remaining $\overline{B}$	Remaining $\overline{A}$ and C_n	C_{n+4}	2
t_{PLH11} / t_{PHL11}	$\overline{B}_i$	None	$\overline{A}_i$	Remaining $\overline{B}$	Remaining $\overline{A}$ and C_n	C_{n+4}	2

1/ In $\overline{A}_i$, $\overline{B}_i$ and $\overline{F}_i$, i = 0, 1, 2, or 3.

FIGURE 5. Waveforms for propagation delay time and test circuit for device type 01 - Continued.

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DIFF MODE TEST TABLE

FUNCTION INPUTS: S1 = S2 = 2.7 V, S0 = S3 = M = 0 V

TEST	INPUT UNDER TEST	OTHER INPUT SAME BIT		OTHER DATA INPUTS		OUTPUT UNDER TEST	OUTPUT WAVEFORM
		APPLY 2.7 V	APPLY GND	APPLY 2.7 V	APPLY GND		
t_{PLH2} / t_{PHL2}	$\overline{A}_i$ 1/	None	$\overline{B}_i$	Remaining $\overline{A}$	Remaining $\overline{B}$, C_n	Any $\overline{F}_i$	1
t_{PLH2} / t_{PHL2}	$\overline{B}_i$	$\overline{A}_i$	None	Remaining $\overline{A}$	Remaining $\overline{B}$, C_n	Any $\overline{F}_i$	2
t_{PLH4} / t_{PHL4}	$\overline{A}_i$	None	$\overline{B}_i$	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	$\overline{P}$	1
t_{PLH4} / t_{PHL4}	$\overline{B}_i$	$\overline{A}_i$	None	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	$\overline{P}$	2
t_{PLH6} / t_{PHL6}	$\overline{A}_i$	$\overline{B}_i$	None	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	$\overline{G}$	1
t_{PLH6} / t_{PHL6}	$\overline{B}_i$	None	$\overline{A}_i$	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	$\overline{G}$	2
t_{PLH8} / t_{PHL8}	$\overline{A}_i$	None	$\overline{B}_i$	Remaining $\overline{A}$	Remaining $\overline{B}$, C_n	A = B	1
t_{PLH8} / t_{PHL8}	$\overline{B}_i$	$\overline{A}_i$	None	Remaining $\overline{A}$	Remaining $\overline{B}$, C_n	A = B	2
t_{PLH10} / t_{PHL10}	C_n	None	None	All $\overline{A}$ and $\overline{B}$	None	C_{n+4}	1
t_{PLH12} / t_{PHL12}	$\overline{A}_i$	$\overline{B}_i$	None	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	C_{n+4}	2
t_{PLH12} / t_{PHL12}	$\overline{B}_i$	None	$\overline{A}_i$	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	C_{n+4}	1

1/ In $\overline{A}_i$, $\overline{B}_i$ and $\overline{F}_i$, i = 0, 1, 2, or 3.

FIGURE 5. Waveforms for propagation delay time and test circuit for device type 01 - Continued.

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LOGIC MODE TEST TABLE

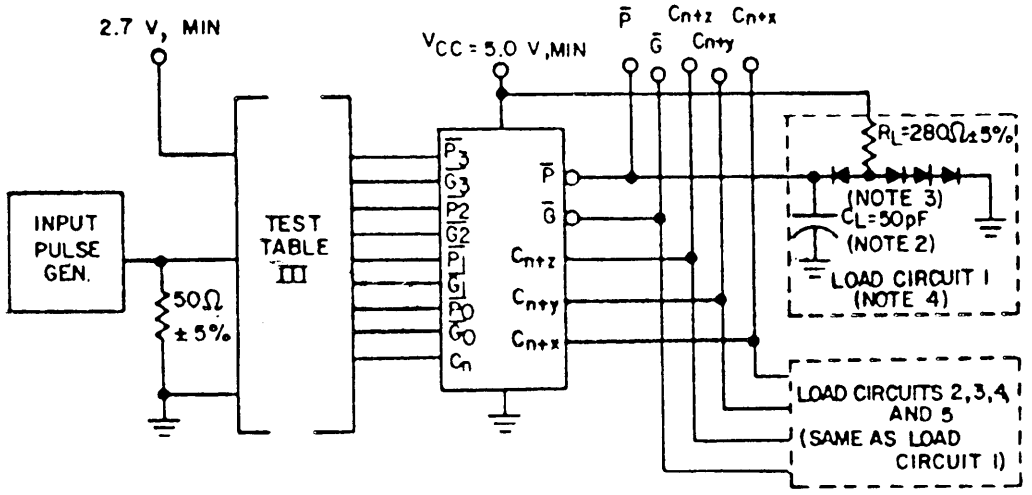
FUNCTION INPUTS: S1 = S2 = M = 2.7 V, S0 = S3 = 0 V

TEST	INPUT UNDER TEST	OTHER INPUT SAME BIT		OTHER DATA INPUTS		OUTPUT UNDER TEST	OUTPUT WAVEFORM
		APPLY 2.7 V	APPLY GND	APPLY 2.7 V	APPLY GND		
t_{PLH13} / t_{PHL13}	$\overline{A}_i$ 1/	None	$\overline{B}_i$	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	Any $\overline{F}_i$	1
t_{PLH13} / t_{PHL13}	$\overline{B}_i$	None	$\overline{A}_i$	None	Remaining $\overline{A}$ and $\overline{B}$, C_n	Any $\overline{F}_i$	1

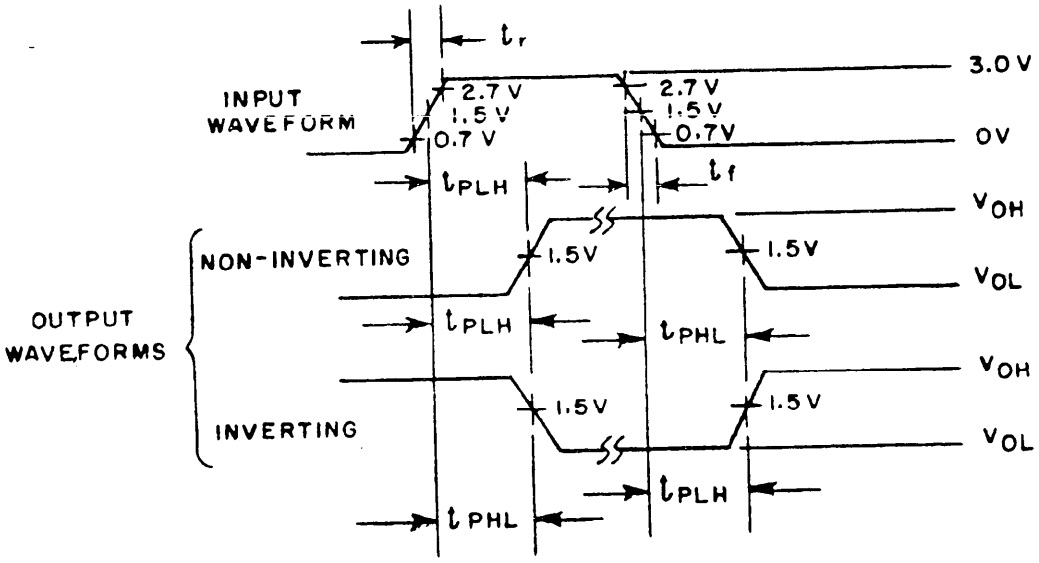
1/ In $\overline{A}_i$, $\overline{B}_i$ and $\overline{F}_i$, i = 0, 1, 2, or 3.

FIGURE 5. Waveforms for propagation delay time and test circuit for device type 01 - Continued.

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TEST CIRCUIT



NOTES:
1. The input pulse has the following characteristics: PRR ≤ 1 MHz, $t_r = t_f \leq 2.5$ ns, $Z_{OUT} \approx 50\Omega$.
2. C_L includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. Load circuit is required on a given output only where table III indicates "OUT" on that output. Load circuits may otherwise be omitted.

FIGURE 5. Waveforms for propagation delay time and test circuit for device type 02 - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J,K,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	$\overline{S}_0$	$\overline{X}_0$	S_3	S_2	S_1	S_0	C_n	M	F_0	F_1	F_2	GND	F_3	A=B	F	C_{n+4}	G	$\overline{S}_3$	X_3	$\overline{S}_2$	X_2	$\overline{S}_1$	X_1	V_{CC}				
T_C = +25°C	V_{OH}	3006	1	0.8 V	2.0 V	2.0 V	0.8 V	0.8 V	0.8 V	2.0 V					GND			-1 mA	-1 mA	-1 mA	0.8 V	2.0 V	0.8 V	2.0 V	0.8 V	2.0 V	4.5 V	$\overline{S}_{n+4}$, $\overline{S}_0$, $\overline{S}_1$, $\overline{S}_2$, $\overline{S}_3$	2.5		V
			2							0.8 V											2.0 V	0.8 V	2.0 V								
			3							2.0 V												2.0 V									
			4																			0.8 V									
			5					2.0 V														2.0 V									
			6						2.0 V		2.0 V	-1 mA	-1 mA	-1 mA		-1 mA															
			7								0.8 V																				
	V_{OL}	3007	8					0.8 V		0.8 V								20 mA	20 mA	20 mA	0.8 V	0.8 V	0.8 V		0.8 V			$\overline{S}_{n+4}$, $\overline{S}_0$, $\overline{S}_1$, $\overline{S}_2$, $\overline{S}_3$, A = B	0.2	0.5	V
			9																												
			10																												
			11																												
			12																												
			13																												
			14																												
			15																												
	I_{IL1}	3009	16	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	0.5 V										5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	M	-1	-2	mA
	I_{IL2}		17	0.5 V																								$\overline{S}_0$, $\overline{S}_1$, $\overline{S}_2$, $\overline{S}_3$	-3	-6	
			18	5.5 V																											
			19																												
			20																		0.5 V		0.5 V		0.5 V						
	I_{IL3}		21						0.5 V												5.5 V							S_0, S_3	-4	-8	
			22			0.5 V			5.5 V												5.5 V										
	I_{IL4}		23	GND		GND		0.5 V		GND	GND										GND		GND		GND			S_1, S_2			
			24				0.5 V	GND		GND																					
	I_{IL5}		25		GND		GND	GND		0.5 V												GND		GND		GND		C_n	-5	-10	
	I_{IL6}		26		0.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V										5.5 V	6.5 V	5.5 V	5.5 V	5.5 V	5.5 V			-2	-4	
			27	5.5 V	0.5 V																5.5 V		5.5 V	0.5 V	GND	0.5 V					
			28																		GND	0.5 V	GND	0.5 V	5.5 V	5.5 V					
			29																				5.5 V		5.5 V	5.5 V					
	I_{IH1}	3010	30	GND	GND	GND	GND	GND	GND	GND	2.7 V											GND	GND	GND	GND	GND		M		50	μA
	I_{IH2}		31		2.7 V						GND													2.7 V		GND		$\overline{S}_0$, $\overline{S}_1$, $\overline{S}_2$, $\overline{S}_3$, M		150	
			32		GND																	2.7 V		GND		2.7 V					
			33																			GND				GND					
			34																												
			35	2.7 V																											
			36	GND																			2.7 V								
			37																				GND								
			38																		2.7 V										
	I_{IH3}		39							2.7 V											GND							S_0, S_3		700	
			40			2.7 V				GND											GND									700	

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J,K,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	$\overline{E}_0$	$\overline{X}_0$	S_3	S_2	S_1	S_0	C_n	M	$\overline{F}_C$	$\overline{F}_1$	$\overline{F}_2$	GND	$\overline{F}_3$	A=B	Y	C_{n+4}	$\overline{G}$	$\overline{E}_3$	X_3	$\overline{E}_2$	X_2	$\overline{E}_1$	X_1	V_{CC}				
1 T_C = +25°C	I_{IH4}	3010	41 42	5.5 V	GND	GND GND	GND 2.7 V	2.7 V GND	GND GND	GND GND	GND GND				GND						5.5 V	GND GND	5.5 V	GND GND	5.5 V	GND GND	5.5 V	S_1 S_2		200 200	µA
	I_{IH5}		43		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	2.7 V	5.5 V											5.5 V		5.5 V		5.5 V		C_n		250	
	I_{IH6}		44	GND	GND	GND	GND	GND	GND	GND	5.5 V										GND	GND	GND	GND	GND	GND		M		1	mA
	I_{IH7}		45 46 47 48 49 50 51 52	5.5 V GND	5.5 V GND						GND										5.5 V	5.5 V GND	5.5 V GND	5.5 V GND	5.5 V GND	GND 5.5 V GND		S_0 S_1 S_2 S_3			
	I_{IH8}		53 54			5.5 V			5.5 V GND												GND GND							S_3			
	I_{IH9}		55 56	5.5 V		GND GND	5.5 V	5.5 V GND													5.5 V		5.5 V		5.5 V			S_1 S_2			
	I_{IH10}		57		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V									GND		5.5 V	GND	GND	GND	GND		$\overline{G}$	-40	-100	
	I_{OS1}	3011	58	GND	GND	5.5 V	GND	GND	GND	GND	GND								GND		GND	GND	GND	GND	GND	GND		C_{n+4}			
	I_{OS2}		59	GND	GND	GND	GND	GND	GND	GND	GND							GND													
	I_{OS3}		60 61 62 63 64	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	GND	GND		GND					5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		S_0 S_1 S_2 S_3			
	I_{CCL}	3005	65	GND		4.5 V	4.5 V	4.5 V	4.5 V	GND	4.5 V										GND	4.5 V	GND	4.5 V	GND	4.5 V		V_{CC}		180	
	I_{CCH}	3005	66	GND	GND	4.5 V	4.5 V	4.5 V	4.5 V	GND	4.5 V										GND	GND	GND	GND	GND	GND		V_{CC}		180	
	V_{IC}		67 68 69 70 71 72 73 74 75 76 77 78 79 80	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA										-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	4.5 V			-1.2	V

See footnotes at end of device type 01.

MIL-M-38510/78A

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	$\overline{P}_0$	$\overline{X}_0$	S_3	S_2	S_1	S_0	C_n	M	F_0	F_1	F_2	GND	F_3	A=B	P	C_{n+4}	G	$\overline{P}_3$	$\overline{X}_3$	$\overline{P}_2$	$\overline{X}_2$	$\overline{P}_1$	$\overline{X}_1$	V_{CC}	A = B S_3 S_2 S_1 S_0 C_{n+4} P G F_3			
1 T_C = +25°C	I_{CEX}		81 82 83 84 85 86 87 88	2.0 V GND 2.0 V " " " " "	2.0 V GND 2.0 V " " " " "	2.0 V " " " " " " "	2.0 V GND 2.0 V " " " " "	2.0 V GND " 2.0 V " " " "	2.0 V GND " " 2.0 V " " "	2.0 V GND " " " 2.0 V " "	2.0 V " " " " " " "	5.5 V " " " " " " "	5.5 V " " " " " " "	5.5 V " " " " " " "	GND " " " " " " "	5.5 V " " " " " " "	5.5 V " " " " " " "	5.5 V " " " " " " "	5.5 V " " " " " " "	5.5 V " " " " " " "	2.0 V " " " " " " "	2.0 V 2.0 V GND 2.0 V GND 2.0 V " "	2.0 V GND 2.0 V " " " " "	2.0 V GND 2.0 V " " " " "	2.0 V GND 2.0 V " " " " "	2.0 V GND 2.0 V " " " " "	4.5 V 5.5 V " " " " " "	A = B S_3 S_2 S_1 S_0 C_{n+4} P G F_3		1 " " " " " " "	mA " " " " " " "
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C, and V_{IC} tests are omitted. V_{IL} = 0.7 V dc and $V_{OL(max)}$ = 0.45 V dc.																														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C, and V_{IC} tests are omitted.																														
7 T_C = +25°C	Functional tests 1/ 4/		89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122	B 2/ " " A B B A B " " " " " " " " " " " " " " " A B A B A B A B B	B 2/ " " A B B A B " " " " " " " " " " " " " " " A B A B A B A B B	B 2/ " " B A A B A " " " " " " " " " " " " " " " B A B A B A B A A	B 2/ " " B A A B A " " " " " " " " " " " " " " " B A B A B A B A A	B 2/ " " A B B A B " " " " " " " " " " " " " " " A B A B A B A B B

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be $\geq$ 2.0 V, $\leq$ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883-method	Cases J,K,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	$\overline{E}_0$	X_0	S_3	S_2	S_1	S_0	C_n	M	F_0	F_1	F_2	GND	F_3	A=B	P	C_{n+4}	G	$\overline{E}_3$	X_3	$\overline{E}_2$	X_2	$\overline{E}_1$	X_1	V_{CC}				
T_C = 9 +25°C	t_{PLH1}	3003 (Fig. 5)	131	2.7 V	IN	2.7 V	GND	GND	2.7 V	GND	GND	OUT	OUT	OUT	GND	OUT					2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	7.7 V	5.0 V	I_{O0} to I_{O1}	2	20	ns
			132	"	2.7 V	"	"	"	"	"	"										"	"	"	"	"	IN	"	I_{O1} to I_{O2}			
			133	"	"	"	"	"	"	"	"										"	"	"	IN	"	2.7 V	"	I_{O2} to I_{O3}			
			134	"	"	"	"	"	"	"	"										"	"	"	"	"	"	"	I_{O3} to I_{O0}			
			135	IN	"	"	"	"	"	"	"										"	"	"	"	"	"	"	I_{O0} to I_{O2}			
			136	2.7 V	"	"	"	"	"	"	"										"	"	"	"	"	"	"	I_{O1} to I_{O3}			
			137	"	"	"	"	"	"	"	"										"	"	"	"	"	"	"	I_{O2} to I_{O0}			
			138	"	"	"	"	"	"	"	"										"	"	"	"	"	"	"	I_{O3} to I_{O1}			
	t_{PHL2}		139	GND	IN	GND	2.7 V	2.7 V	GND			OUT	OUT	OUT		OUT					GND		GND		GND	IN		I_{O0} to I_{O1}		25	
			140	"	2.7 V	"	"	"	"												"	IN	"	IN	"	2.7 V		I_{O1} to I_{O2}			
			141	"	"	"	"	"	"												"	2.7 V	"	2.7 V	"	"		I_{O2} to I_{O3}			
			142	"	"	"	"	"	"												"	"	"	"	"	"		I_{O3} to I_{O0}			
			143	IN	"	"	"	"	"												"	"	"	"	"	"		I_{O0} to I_{O2}			
			144	GND	"	"	"	"	"												"	"	"	"	"	"		I_{O1} to I_{O3}			
			145	"	"	"	"	"	"												"	"	"	"	"	"		I_{O2} to I_{O0}			
			146	"	"	"	"	"	"												"	"	"	"	"	"		I_{O3} to I_{O1}			
	t_{PLH2}		147		IN							OUT	OUT	OUT							GND							I_{O0} to I_{O1}		23	
			148		2.7 V																"			IN		IN		I_{O1} to I_{O2}			
			149		"																"			2.7 V		2.7 V		I_{O2} to I_{O3}			
			150		"																"			"		"		I_{O3} to I_{O0}			
			151	IN	"																"			"		"		I_{O0} to I_{O2}			
			152	GND	"																"			"		"		I_{O1} to I_{O3}			
			153	"	"																"			"		"		I_{O2} to I_{O0}			
			154	"	"																"			"		"		I_{O3} to I_{O1}			
	t_{PHL3}		155	2.7 V	IN	2.7 V	GND	GND	2.7 V									OUT			GND	GND	2.7 V	GND	2.7 V	GND		I_{O0} to I_{O1}		14	
			156	GND	GND																"	IN	GND	IN	"	IN		I_{O1} to I_{O2}			
			157	"	"																2.7 V	GND	"	GND	"	GND		I_{O2} to I_{O3}			
			158	"	"																GND	"	"	"	"	"		I_{O3} to I_{O0}			
			159	IN	2.7 V																"	IN	IN	"	IN	"		I_{O0} to I_{O2}			
			160	GND	GND																"	GND	"	2.7 V	"	2.7 V		I_{O1} to I_{O3}			
			161	"	"																"	"	IN	"	IN	"		I_{O2} to I_{O0}			
			162	"	"																IN	2.7 V	GND	GND	GND	GND		I_{O3} to I_{O1}			
	t_{PLH3}		163	2.7 V	IN													OUT			GND	GND	IN	GND	IN	IN		I_{O0} to I_{O1}			
			164	GND	GND																"	IN	GND	IN	GND	IN		I_{O1} to I_{O2}			
			165	"	"																2.7 V	GND	"	GND	"	GND		I_{O2} to I_{O3}			
			166	"	"																GND	"	"	"	"	"		I_{O3} to I_{O0}			
			167	IN	2.7 V																"	IN	IN	"	IN	"		I_{O0} to I_{O2}			
			168	GND	GND																"	GND	"	2.7 V	"	2.7 V		I_{O1} to I_{O3}			
			169	"	"																"	"	IN	"	IN	"		I_{O2} to I_{O0}			
			170	"	"																IN	2.7 V	GND	GND	GND	GND		I_{O3} to I_{O1}			
	t_{PHL4}		171		IN	GND	2.7 V	2.7 V	GND												GND							I_{O0} to I_{O1}		18	
			172		GND	"	"	"	"												"	IN		IN		IN		I_{O1} to I_{O2}			
			173		"	"	"	"	"												"	GND		GND		GND		I_{O2} to I_{O3}			
			174		"	"	"	"	"												"	"		"		"		I_{O3} to I_{O0}			
			175	IN	"	"	"	"	"												"	IN		IN		IN		I_{O0} to I_{O2}			
			176	GND	"	"	"	"	"												"	GND		GND		GND		I_{O1} to I_{O3}			
			177	"	"	"	"	"	"												"	"		"		"		I_{O2} to I_{O0}			
			178	"	"	"	"	"	"												"	"		"		"		I_{O3} to I_{O1}			

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J,K,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test Limits Min	Test Limits Max	Unit
			Test no.	B_0	A_0	S_3	S_2	S_1	S_0	C_n	M	F_0	F_1	F_2	GND	F_3	A-B	F	C_{n+4}	$\overline{S}$	B_3	A_3	B_2	A_2	B_1	A_1	V_{CC}				
9 T_C = +25°C	t_{PLH4}	3003 (Fig. 5)	179	GND	IN	GND	2.7 V	2.7 V	GND	GND	GND				GND			OUT			GND	GND	GND	GND	GND	5.0 V	$\overline{F}_0$	2	18	ns
			180		GND																			IN		IN		$\overline{F}_1$			
			181																					GND		GND		$\overline{F}_2$			
			182																									$\overline{F}_3$			
			183	IN	2.7 V																	IN				2.7 V		$\overline{A=B}$			
			184	GND	GND																	GND		IN		GND					
			185																					GND							
			186																		IN	2.7 V	GND								
	t_{PHL5}		187		IN	2.7 V	GND	GND	2.7 V									OUT		OUT	2.7 V	GND	2.7 V		2.7 V			$\overline{F}_0$		14	
			188	2.7 V	GND																		2.7 V	IN	GND			$\overline{F}_1$			
			189																		GND	IN	GND			IN		$\overline{F}_2$			
			190																		2.7 V	GND	2.7 V		2.7 V			$\overline{F}_3$			
			191	IN																			GND					$\overline{A=B}$			
			192	2.7 V																			2.7 V		IN						
			193																		IN	2.7 V			2.7 V						
			194																												
	t_{PLH5}		195	GND	IN																2.7 V							$\overline{F}_0$		14	
			196	2.7 V	GND																		GND	IN	GND			$\overline{F}_1$			
			197																		GND	IN	2.7 V			IN		$\overline{F}_2$			
			198																		2.7 V	GND	2.7 V		2.7 V			$\overline{F}_3$			
			199	IN																			GND					$\overline{A=B}$			
			200	2.7 V																			2.7 V		IN						
			201																		IN	2.7 V			2.7 V						
			202																												
	t_{PHL6}		203		IN	GND	2.7 V	2.7 V	GND												GND	GND	GND	GND	GND	5.0 V	$\overline{F}_0$		18	
			204	GND	GND																GND	GND	2.7 V	IN	2.7 V	IN		$\overline{F}_1$			
			205																		2.7 V	IN	GND	GND	GND	GND		$\overline{F}_2$			
			206																		GND	GND	2.7 V	IN	2.7 V	IN		$\overline{F}_3$			
			207	IN																	2.7 V	IN	GND	GND	GND	GND		$\overline{A=B}$			
			208	GND																	GND	GND	2.7 V	IN	2.7 V	IN					
			209																		IN	IN	GND	GND	GND	GND					
			210																												
	t_{PHL6}		211	2.7 V	IN																GND							$\overline{F}_0$		18	
			212	GND	GND																GND	IN	2.7 V	IN	2.7 V	IN		$\overline{F}_1$			
			213																		2.7 V	IN	GND	GND	GND	GND		$\overline{F}_2$			
			214																		GND	GND	2.7 V	IN	2.7 V	IN		$\overline{F}_3$			
			215	IN																	2.7 V	IN	GND	GND	GND	GND		$\overline{A=B}$			
			216	GND																	GND	GND	2.7 V	IN	2.7 V	IN					
			217																		IN	IN	GND	GND	GND	GND					
			218																												
	t_{PHL7}	*	219	*	2.7 V	2.7 V	GND	GND	2.7 V	IN	*	OUT			*						GND	2.7 V	*	2.7 V	*	2.7 V	*	C_n to $\overline{F}_0$	*	14	*
	t_{PLH7}	*	220	*	2.7 V	2.7 V	GND	GND	2.7 V	IN	*	OUT			*						*	*	*	*	*	*	*	C_n to $\overline{F}_0$	*	14	*
	t_{PHL8}		221		IN	GND	2.7 V	2.7 V	GND	GND							OUT											A to A=B		33	
			222		2.7 V																			IN		IN		B to A=B			
			223																					2.7 V		2.7 V		C to A=B			
			224																									D to A=B			
			225	IN																		IN			IN			$\overline{A=B}$ to A=B			
			226	GND																		2.7 V			GND						
			227																		IN				GND						
			228																		GND										

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	Test No.	1 $\overline{E}_0$	2 A_0	3 S_3	4 S_2	5 S_1	6 S_0	7 C_n	8 M	9 F_0	10 F_1	11 F_2	12 GND	13 F_3	14 A=B	15 P	16 C_{n+4}	17 G	18 B_3	19 X_3	20 B_2	21 X_2	22 B_1	23 X_1	24 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
T_C = +25°C	t_{PLH9}	3003 (Fig. 5)		229	GND	I/O 2.7 V	GND	2.7 V	2.7 V	GND	GND	GND				GND		XXX			GND	2.7 V	GND	2.7 V	GND	2.7 V	5.0 V	to A=B	2	27	ns
				230	2.7 V																										
				231																											
				232																											
				233																											
				234																											
				235																											
				236	GND																										
	t_{PHL9}			237			2.7 V	GND	GND	2.7 V	I/O									CUT								C_n to C_{n+4}			
	t_{PLH8}			238			2.7 V	GND	GND	2.7 V																		C_n to C_{n+4}			
	t_{PHL8}			239	2.7 V		GND	2.7 V	2.7 V	GND																		C_n to C_{n+4}			
	t_{PLH10}			240	2.7 V		GND	2.7 V	2.7 V	GND																		C_n to C_{n+4}			
	t_{PHL10}			241	GND	I/O			GND		GND																	to C_{n+4}			
				242	2.7 V	GND																									
				243																											
				244																											
				245	I/O																										
				246	2.7 V																										
				247																											
				248																											
	t_{PLH11}			249	GND	I/O																									
				250	2.7 V	GND																									
				251																											
				252																											
				253																											
				254																											
				255																											
				256	I/O																										
	t_{PHL12}			257		I/O	GND																								
				258	GND	GND																									
				259																											
				260																											
				261																											
				262																											
				263																											
				264		I/O	GND																								
	t_{PLH12}			265																											
				266																											
				267																											
				268																											
				269																											
				270																											
				271																											
				272																											

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TABLE II:. Group A inspection for devi:e type 01 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883-method	Cases J,K,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	$\overline{P}_0$	X_0	S_3	S_2	S_1	S_0	C_n	M	F_0	F_1	F_2	GND	F_3	A=B	P	C_{n+4}	G	$\overline{P}_3$	X_3	$\overline{P}_2$	X_2	$\overline{P}_1$	X_1	V_{CC}				
9 T_C = +25°C	t_{PHL13}	3003 (Fig. 5)	273	GND	IN	GND	2.7 V	2.7 V	GND	GND	2.7 V	OUT	CUT	OUT	GND	OUT					GND	GND	GND	GND	GND	GND	5.0 V	X_0 to F_0	2	26	ns
			274	"	GND	"	"	"	"	"	"	"	CUT	"	"	"						"	"	"	IN	"	IN	"	X_1 to F_1	"	"	
			275	"	"	"	"	"	"	"	"	"	"	OUT	"	"						"	IN	"	GND	"	GND	"	X_2 to F_2	"	"	
			276	"	"	"	"	"	"	"	"	"	"	"	"	"						"	GND	"	"	"	"	"	X_3 to F_3	"	"	
			277	IN	"	"	"	"	"	"	"	OLT	OUT	"	"							"	"	"	"	"	"	"	$\overline{P}_0$ to F_0	"	"	
			278	GND	"	"	"	"	"	"	"	"	"	"	"							"	"	IN	"	IN	"	"	$\overline{P}_1$ to F_1	"	"	
			279	"	"	"	"	"	"	"	"	"	"	"	"							"	"	GND	"	GND	"	"	$\overline{P}_2$ to F_2	"	"	
			280	"	"	"	"	"	"	"	"	"	"	"	"							IN	"	"	"	"	"	"	$\overline{P}_3$ to F_3	"	"	
	t_{PLH13}		281	"	IN	"	"	"	"	"	"	OLT	OUT	"	"						GND	"	"	"	"	IN	"	X_0 to F_0	"	24	"
			282	"	GND	"	"	"	"	"	"	"	"	"	"						"	"	"	IN	"	GND	"	X_1 to F_1	"	"	
			283	"	"	"	"	"	"	"	"	"	"	"	"						"	"	IN	"	"	"	"	X_2 to F_2	"	"	
			284	"	"	"	"	"	"	"	"	"	"	"	"						"	"	GND	"	"	"	"	X_3 to F_3	"	"	
			285	IN	"	"	"	"	"	"	"	OUT	"	"	"							"	"	"	IN	"	"	"	$\overline{P}_0$ to F_0	"	"	
			286	GND	"	"	"	"	"	"	"	"	"	"	"							"	"	IN	"	GND	"	"	$\overline{P}_1$ to F_1	"	"	
			287	"	"	"	"	"	"	"	"	"	"	"	"							"	"	GND	"	GND	"	"	$\overline{P}_2$ to F_2	"	"	
			288	"	"	"	"	"	"	"	"	"	"	"	"							IN	"	"	"	"	"	"	$\overline{P}_3$ to F_3	"	"	
10 T_C = +125°C	t_{PHL1}		289	2.7 V	IN	2.7 V	GND	GND	2.7 V	"	GND	OUT	OUT	OUT	"	OUT					2.7 V	"	2.7 V	2.7 V	2.7 V	2.7 V	"	X_0 to F_0	"	23	"
			290	"	2.7 V	"	"	"	"	"	"	"	"	"	"	"						"	"	"	IN	"	IN	"	X_1 to F_1	"	"	
			291	"	"	"	"	"	"	"	"	"	"	"	"	"						"	IN	"	2.7 V	"	2.7 V	"	X_2 to F_2	"	"	
			292	"	"	"	"	"	"	"	"	"	"	"	"	"						"	2.7 V	"	"	"	"	"	X_3 to F_3	"	"	
			293	IN	"	"	"	"	"	"	"	OUT	"	"	"							"	"	"	"	"	"	"	$\overline{P}_0$ to F_0	"	"	
			294	2.7 V	"	"	"	"	"	"	"	"	"	"	"							"	"	IN	"	IN	"	"	$\overline{P}_1$ to F_1	"	"	
			295	"	"	"	"	"	"	"	"	"	"	"	"							"	"	2.7 V	"	2.7 V	"	"	$\overline{P}_2$ to F_2	"	"	
			296	"	"	"	"	"	"	"	"	"	"	"	"							IN	"	"	"	"	"	"	$\overline{P}_3$ to F_3	"	"	
	t_{PLH1}		297	"	IN	"	"	"	"	"	"	OUT	OUT	OUT	"	OUT					2.7 V	"	"	"	"	IN	"	X_0 to F_0	"	"	"
			298	"	2.7 V	"	"	"	"	"	"	"	"	"	"	"						"	IN	"	IN	"	2.7 V	"	X_1 to F_1	"	"	
			299	"	"	"	"	"	"	"	"	"	"	"	"	"						"	2.7 V	"	"	"	"	"	X_2 to F_2	"	"	
			300	"	"	"	"	"	"	"	"	"	"	"	"	"						"	"	"	"	"	"	"	X_3 to F_3	"	"	
			301	IN	"	"	"	"	"	"	"	OUT	"	"	"							"	"	"	"	"	"	"	$\overline{P}_0$ to F_0	"	"	
			302	2.7 V	"	"	"	"	"	"	"	"	"	"	"							"	"	IN	"	IN	"	"	$\overline{P}_1$ to F_1	"	"	
			303	"	"	"	"	"	"	"	"	"	"	"	"							"	"	2.7 V	"	2.7 V	"	"	$\overline{P}_2$ to F_2	"	"	
			304	"	"	"	"	"	"	"	"	"	"	"	"							IN	"	"	"	"	"	"	$\overline{P}_3$ to F_3	"	"	
	t_{PHL2}		305	GND	IN	GND	2.7 V	2.7 V	GND	"	"	OUT	OUT	OUT	"	OUT					GND	"	GND	"	GND	"	"	X_0 to F_0	"	28	"
			306	"	2.7 V	"	"	"	"	"	"	"	"	"	"	"						"	"	"	IN	"	IN	"	X_1 to F_1	"	"	
			307	"	"	"	"	"	"	"	"	"	"	"	"	"						"	IN	"	2.7 V	"	2.7 V	"	X_2 to F_2	"	"	
			308	"	"	"	"	"	"	"	"	"	"	"	"	"						"	2.7 V	"	"	"	"	"	X_3 to F_3	"	"	
			309	IN	"	"	"	"	"	"	"	OUT	"	"	"							"	"	"	"	"	"	"	$\overline{P}_0$ to F_0	"	"	
			310	GND	"	"	"	"	"	"	"	"	"	"	"							"	"	IN	"	IN	"	"	$\overline{P}_1$ to F_1	"	"	
			311	"	"	"	"	"	"	"	"	"	"	"	"							"	"	GND	"	GND	"	"	$\overline{P}_2$ to F_2	"	"	
			312	"	"	"	"	"	"	"	"	"	"	"	"							IN	"	"	"	"	"	"	$\overline{P}_3$ to F_3	"	"	
	t_{PLH2}		313	"	IN	"	"	"	"	"	"	OUT	OUT	OUT	"	OUT					GND	"	"	"	"	IN	"	X_0 to F_0	"	26	"
			314	"	2.7 V	"	"	"	"	"	"	"	"	"	"	"						"	IN	"	IN	"	2.7 V	"	X_1 to F_1	"	"	
			315	"	"	"	"	"	"	"	"	"	"	"	"	"						"	2.7 V	"	"	"	"	"	X_2 to F_2	"	"	
			316	"	"	"	"	"	"	"	"	"	"	"	"	"						"	"	"	"	"	"	"	X_3 to F_3	"	"	
			317	IN	"	"	"	"	"	"	"	OUT	"	"	"							"	"	"	"	"	"	"	$\overline{P}_0$ to F_0	"	"	
			318	GND	"	"	"	"	"	"	"	"	"	"	"							"	"	IN	"	IN	"	"	$\overline{P}_1$ to F_1	"	"	
			319	"	"	"	"	"	"	"	"	"	"	"	"							"	"	GND	"	GND	"	"	$\overline{P}_2$ to F_2	"	"	
			320	"	"	"	"	"	"	"	"	"	"	"	"							IN	"	"	"	"	"	"	$\overline{P}_3$ to F_3	"	"	

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H $\geq$ 2.0 V, L $\leq$ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J,K,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	$\overline{W}_0$	$\overline{X}_0$	S_3	S_2	S_1	S_0	C_n	M	F_0	F_1	F_2	GND	F_3	4-8	$\overline{F}$	C_{n+4}	G	$\overline{W}_3$	$\overline{X}_3$	$\overline{W}_2$	$\overline{X}_2$	$\overline{W}_1$	$\overline{X}_1$	V_{CC}				
10 T_C = +125°C	t_{PHL3}	3003 (Fig. 5)	321 322 323 324 325 326 327 328	2.7 V GND " " IN GND "	IN GND " " 2.7 V GND "	2.7 V "	GND "	GND "	2.7 V "	GND "	GND "				GND "			OUT "			GND " 2.7 V GND " IN	GND " IN GND " 2.7 V	GND GND 2.7 V GND " IN GND	GND GND IN GND " 2.7 V GND	GND 2.7 V GND " IN GND	GND IN GND " 2.7 V GND	5.0 V "	$\overline{W}_3$ to $\overline{X}_3$ $\overline{X}_3$ to $\overline{W}_2$ $\overline{W}_2$ to $\overline{X}_2$ $\overline{X}_2$ to $\overline{W}_1$ $\overline{W}_1$ to $\overline{X}_1$ $\overline{X}_1$ to $\overline{W}_3$	2 "	16 "	ns "
	t_{PLH3}	"	329 330 331 332 333 334 335 336	2.7 V GND " " IN GND "	IN GND " " 2.7 V GND "	"	"	"	"	"	"				"			"			GND " 2.7 V GND " IN	GND " IN GND " 2.7 V	" " 2.7 V GND " IN GND	" IN GND " 2.7 V GND	" 2.7 V GND " IN GND	" IN GND " 2.7 V GND	"	"	"	"
	t_{PHL4}	"	337 338 339 340 341 342 343 344	" " " " IN GND "	IN GND " " 2.7 V GND "	GND "	2.7 V "	2.7 V "	GND "	"	"				"			"			GND " " " " IN	GND " IN GND " 2.7 V	" " " " IN GND	" " IN GND " 2.7 V GND	" " " IN GND	" IN GND " 2.7 V GND	"	"	"	"
	t_{PLH4}	"	345 346 347 348 349 350 351 352	" " " " IN GND "	IN GND " " 2.7 V GND "	"	"	"	"	"	"				"			"			GND " " " " IN	GND " IN GND " 2.7 V	" " " " IN GND	" " IN GND " 2.7 V GND	" " " IN GND GND	" IN GND " 2.7 V GND	"	"	"	"
	t_{PHL5}	"	353 354 355 356 357 358 359 360	" 2.7 V " " IN 2.7 V "	IN GND " " " " "	2.7 V "	GND "	GND "	2.7 V "	"	"				"			"		OUT "	2.7 V " " GND 2.7 V " IN	GND " " IN GND " "	2.7 V 2.7 V GND 2.7 V " IN 2.7 V	" " IN GND " "	2.7 V GND 2.7 V " IN 2.7 V	" IN GND " " "	"	"	"	"
	t_{PLH5}	"	361 362 363 364 365 366 367 368	GND 2.7 V " " IN 2.7 V "	IN GND " " " " "	"	"	"	"	"	"				"			"		"	2.7 V " GND 2.7 V " IN 2.7 V	" " IN GND " "	" " GND 2.7 V " IN 2.7 V	" " IN GND " "	" GND 2.7 V " IN 2.7 V	" IN GND " " "	"	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases U,X,2 Test No.	1 $\overline{E}_0$	2 K_0	3 S_3	4 S_2	5 S_1	6 S_0	7 C_n	8 M	9 F_0	10 F_1	11 F_2	12 GND	13 F_3	14 A-B	15 F	16 C_{n+4}	17 G	18 $\overline{E}_3$	19 K_3	20 $\overline{E}_2$	21 K_2	22 $\overline{E}_1$	23 K_1	24 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Units
10 T_C = +125°C	t_{PHL6}	3003 (F_0, $\overline{S}$)	369 370 371 372 373 374 375 376	2.7 V GND	IN GND	GND	2.7 V	2.7 V	GND	GND	GND				GND					OUT	GND IN GND	GND IN GND	GND 2.7 V GND IN GND	GND GND IN GND	GND 2.7 V GND IN GND	GND IN GND	5.0 V	$\overline{E}_0$ to A-B; K_0 to A-B; S_3 to A-B; S_2 to A-B; S_1 to A-B; S_0 to A-B; C_n to A-B; M to A-B; F_0 to A-B; F_1 to A-B; F_2 to A-B; GND to A-B; F_3 to A-B	2	20	ns
	t_{PLH6}		377 378 379 380 381 382 383 384	2.7 V GND	IN GND																GND 2.7 V GND								$\overline{E}_0$ to A-B; K_0 to A-B; S_3 to A-B; S_2 to A-B; S_1 to A-B; S_0 to A-B; C_n to A-B; M to A-B; F_0 to A-B; F_1 to A-B; F_2 to A-B; GND to A-B; F_3 to A-B			
	t_{PHL7}		385		2.7 V	2.7 V	GND	GND	2.7 V	IN		OUT									GND	2.7 V						C_n to F_0		16	
	t_{PLH7}		386		2.7 V	2.7 V	GND	GND	2.7 V	IN		OUT									GND							C_n to F_0		16	
	t_{PHL8}		387 388 389 390 391 392 393 394		IN 2.7 V	GND	2.7 V	2.7 V	GND	GND						CUT						IN 2.7 V						$\overline{E}_0$ to A-B; K_0 to A-B; S_3 to A-B; S_2 to A-B; S_1 to A-B; S_0 to A-B; C_n to A-B; M to A-B; F_0 to A-B; F_1 to A-B; F_2 to A-B; GND to A-B; F_3 to A-B			
	t_{PLH8}		395 396 397 398 399 400 401 402		IN 2.7 V																							$\overline{E}_0$ to A-B; K_0 to A-B; S_3 to A-B; S_2 to A-B; S_1 to A-B; S_0 to A-B; C_n to A-B; M to A-B; F_0 to A-B; F_1 to A-B; F_2 to A-B; GND to A-B; F_3 to A-B			
	t_{PHL9}		403		2.7 V	2.7 V	GND	GND	2.7 V	IN		OUT									GND							C_n to C_{n+4}		14.5	
	t_{PLH9}		404		2.7 V	2.7 V	GND	GND	2.7 V	IN		OUT									GND							C_n to C_{n+4}		16	
	t_{PHL10}		405		2.7 V	GND	2.7 V	2.7 V	GND												2.7 V		2.7 V					C_n to C_{n+4}		36	
	t_{PLH10}		406		2.7 V	GND	2.7 V	2.7 V	GND												GND		GND					C_n to C_{n+4}		25	
	t_{PHL11}		407 408 409 410 411 412 413 414		GND 2.7 V	2.7 V GND				GND																		C_n to C_{n+4}			
	t_{PLH11}				IN GND	IN GND																									

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883-method	Cases J,K,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	$\overline{W}_0$	$\overline{X}_0$	S_3	S_2	S_1	S_0	C_n	M	F_0	F_1	F_2	GND	F_3	A=B	$\overline{F}$	C_{n+4}	G	$\overline{W}_3$	$\overline{X}_3$	$\overline{W}_2$	$\overline{X}_2$	$\overline{W}_1$	$\overline{X}_1$	V_{CC}				
10 T_C = +125°C.	t_{PLH11}	3003 (Fig. 5)	415 416 417 418 419 420 421 422	GND 2.7 V " " IN 2.7 V "	IN GND " " " " " "	2.7 V " " " " " " "	GND " " " " " " "	GND " " " " " " "	2.7 V " " " " " " "	GND " " " " " " "	GND " " " " " " "				GND " " " " " " "				OUT " " " " " " "		2.7 V " " GND 2.7 V " IN	GND " " IN GND " " "	2.7 V 2.7 V GND 2.7 V " IN 2.7 V	GND " IN GND " " "	2.7 V GND 2.7 V " IN 2.7 V 2.7 V	GND IN GND " " "	5.0 V " " " " "	$\overline{X}_0$ to C_{n+4} $\overline{X}_1$ to C_{n+4} $\overline{X}_2$ to C_{n+4} $\overline{X}_3$ to C_{n+4} $\overline{W}_0$ to C_{n+4} $\overline{W}_1$ to C_{n+4} $\overline{W}_2$ to C_{n+4} $\overline{W}_3$ to C_{n+4}	2 " " " " " " "	25 " " " " " " "	ns " " " " " " "
	t_{PHL12}	" " " " " " "	423 424 425 426 427 428 429 430	" GND " " IN GND " "	IN GND " " " " " "	GND " " " " " " "	2.7 V " " " " " " "	2.7 V " " " " " " "	GND " " " " " " "	" " " " " " "	" " " " " " "				" " " " " " "				" " " " " " "		GND " " 2.7 V GND " IN	" " IN GND " " "	GND GND 2.7 V GND " IN GND	" " IN GND " " "	GND 2.7 V GND " " IN GND	IN GND " " " " "	" " " " " " "	$\overline{X}_0$ to C_{n+4} $\overline{X}_1$ to C_{n+4} $\overline{X}_2$ to C_{n+4} $\overline{X}_3$ to C_{n+4} $\overline{W}_0$ to C_{n+4} $\overline{W}_1$ to C_{n+4} $\overline{W}_2$ to C_{n+4} $\overline{W}_3$ to C_{n+4}	" " " " " " "	29 " " " " " " "	" " " " " " "
	t_{PLH12}	" " " " " " "	431 432 433 434 435 436 437 438	2.7 V GND " " IN GND " "	IN GND " " " " " "	" " " " " " "	" " " " " " "	" " " " " " "	" " " " " " "	" " " " " " "	" " " " " " "				" " " " " " "				" " " " " " "		GND " " 2.7 V GND " IN	" " IN GND " " "	" " 2.7 V GND " IN GND	" " IN GND " " "	" 2.7 V GND " IN GND	" IN GND " " "	" " " " " " "	$\overline{X}_0$ to C_{n+4} $\overline{X}_1$ to C_{n+4} $\overline{X}_2$ to C_{n+4} $\overline{X}_3$ to C_{n+4} $\overline{W}_0$ to C_{n+4} $\overline{W}_1$ to C_{n+4} $\overline{W}_2$ to C_{n+4} $\overline{W}_3$ to C_{n+4}	" " " " " " "	" " " " " " "	" " " " " " "
	t_{PHL13}	" " " " " " "	439 440 441 442 443 444 445 446	" " " " IN GND " "	IN GND " " " " " "	" " " " " " "	" " " " " " "	" " " " " " "	" " " " " " "	" " " " " " "	2.7 V " " " " " " "	OUT OJT	OUT OUT	OUT OUT	OUT " " " " " " "	OUT OUT				GND " " " " " " IN	" " IN GND " " "	" " " " " IN GND	" IN GND " " "	" " " " " IN GND	IN GND " " " " "	" " " " " " "	$\overline{X}_0$ to C_{n+4} $\overline{X}_1$ to C_{n+4} $\overline{X}_2$ to C_{n+4} $\overline{X}_3$ to C_{n+4} $\overline{W}_0$ to F_0 $\overline{W}_1$ to F_1 $\overline{W}_2$ to F_2 $\overline{W}_3$ to F_3	" " " " " " "	28 " " " " " " "	" " " " " " "
	t_{PLH13}	" " " " " " "	447 448 449 450 451 452 453 454	" " " " IN GND " "	IN GND " " " " " "	" " " " " " "	" " " " " " "	" " " " " " "	" " " " " " "	" " " " " " "	" " " " " " "	OUT OUT	OUT OUT	OUT OUT	OUT " " " " " " "	OUT OUT				GND " " " " " " IN	" " IN GND " " "	" " " " " IN GND	" IN GND " " "	" " " " " IN GND	IN GND " " " " "	" " " " " " "	$\overline{X}_0$ to C_{n+4} $\overline{X}_1$ to C_{n+4} $\overline{X}_2$ to C_{n+4} $\overline{X}_3$ to C_{n+4} $\overline{W}_0$ to F_0 $\overline{W}_1$ to F_1 $\overline{W}_2$ to F_2 $\overline{W}_3$ to F_3	" " " " " " "	26 " " " " " " "	" " " " " " "
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																														

1/ Tests shall be performed in sequence.

2/ Inputs:
A = 3.0 V minimum
B = 0.0 V or GND

3/ Outputs:
H > 1.5 V
L < 1.5 V

4/ Only a summary of attributes data is required.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test no.	1 $\overline{G}_1$	2 $\overline{P}_1$	3 $\overline{G}_0$	4 $\overline{P}_0$	5 $\overline{G}_3$	6 $\overline{P}_3$	7 $\overline{P}$	8 GND	9 C_{n+Z}	10 $\overline{G}$	11 C_{n+Y}	12 C_{n+X}	13 C_n	14 $\overline{G}_2$	15 $\overline{P}_2$	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
T_C +25°C	V_{OH}	3006	1		0.8 V		2.0 V		0.8 V	-1 mA	GND	-1 mA	-1 mA			0.8 V	0.8 V	0.8 V	4.5 V	$\overline{P}$	2.5		V
		"	2		2.0 V		2.0 V		2.0 V		"						0.8 V	2.0 V	"	C_{n+Z}	"		"
		"	3	2.0 V	0.8 V						"								"	$\overline{G}$	"		"
		"	4	0.8 V	0.8 V	2.0 V	2.0 V	2.0 V			"			-1 mA		0.8 V		0.8 V	"	C_{n+Y}	"		"
		"	5		2.0 V	0.8 V	2.0 V		2.0 V		"				-1 mA	0.8 V			"	C_{n+X}	"		"
	V_{OL}	3007	6		0.8 V		0.8 V		0.8 V	20 mA	"	20 mA	20 mA			2.0 V	2.0 V	0.8 V	"	$\overline{P}$	0.2	0.5	"
		"	7	0.8 V	0.8 V	0.8 V	0.8 V				"						2.0 V	2.0 V	"	C_{n+Z}		"	"
		"	8	2.0 V	2.0 V	2.0 V		0.8 V	2.0 V		"			20 mA		2.0 V			"	$\overline{G}$		"	"
		"	9	2.0 V	2.0 V	0.8 V					"				20 mA	2.0 V			"	C_{n+Y}		"	"
		"	10			2.0 V	2.0 V				"								"	C_{n+X}		"	"
	V_{IC}		11	-18 mA							"								"	$\overline{G}_1$		-1.2	"
			12		-18 mA						"								"	$\overline{P}_1$		"	"
			13			-18 mA					"								"	$\overline{G}_0$		"	"
			14				-18 mA				"								"	$\overline{P}_0$		"	"
			15					-18 mA			"								"	$\overline{G}_3$		"	"
			16						-18 mA		"								"	$\overline{P}_3$		"	"
			17								"								"	C_n		"	"
			18								"								"	$\overline{G}_2$		"	"
			19								"								"	$\overline{P}_2$		"	"
	I_{IL1}	3009	20	0.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		"					GND	5.5 V	5.5 V	5.5 V	$\overline{G}_1$	-8	-16	mA
	I_{IL2}	"	21	5.5 V	0.5 V	5.5 V	"	"	"		"					"	"	"	"	$\overline{P}_1$	-4	-8	"
	I_{IL3}	"	22	"	5.5 V	0.5 V	"	"	"		"					"	"	"	"	$\overline{G}_0$	-7	-14	"
	I_{IL4}	"	23	"	"	5.5 V	0.5 V	"	"		"					5.5 V	"	"	"	$\overline{P}_0$	-4	-8	"
	I_{IL5}	"	24	"	"	"	5.5 V	0.5 V	"		"					"	"	"	"	$\overline{G}_3$	-4	-8	"
	I_{IL6}	"	25	"	"	"	"	5.5 V	0.5 V		"					"	"	"	"	$\overline{P}_3$	-2	-4	"
	I_{IL7}	"	26	"	"	"	"	"	5.5 V		"					0.5 V	"	"	"	C_n	-1	-2	"
	I_{IL8}	"	27	"	"	"	"	"	"		"					GND	0.5 V	"	"	$\overline{G}_2$	-7	-14	"
	I_{IL9}	"	28	"	"	"	"	"	"		"					5.5 V	5.5 V	0.5 V	"	$\overline{P}_2$	-3	-6	"
	I_{IH1}	3010	29	2.7 V	GND	GND	GND	GND	GND		"					"	GND	GND	"	$\overline{G}_1$		400	μA
	I_{IH2}	"	30	GND	2.7 V	GND	"	"	"		"					"	"	"	"	$\overline{P}_1$		200	"
	I_{IH3}	"	31	GND	GND	2.7 V	"	"	"		"					"	"	"	"	$\overline{G}_0$		350	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{G}_1$	$\overline{P}_1$	$\overline{G}_0$	$\overline{P}_0$	$\overline{G}_3$	$\overline{P}_3$	$\overline{P}$	GND	C_{n+Z}	$\overline{G}$	C_{n+Y}	C_{n+X}	C_n	$\overline{G}_2$	$\overline{P}_2$	V_{CC}				
1 T_C +25°C	I_{IH4}	3010	32	GND	GND	GND	2.7 V	GND	GND		GND					GND	GND	GND	5.5 V	$\overline{P}_0$		200	µA
	I_{IH5}	"	33	"	"	"	GND	2.7 V	GND		"					"	"	"	"	$\overline{G}_3$		200	"
	I_{IH6}	"	34	"	"	"	"	GND	2.7 V		"					"	"	"	"	$\overline{P}_3$		100	"
	I_{IH7}	"	35	"	"	"	"	"	GND		"					2.7 V	"	"	"	C_n		50	"
	I_{IH8}	"	36	"	"	"	"	"	"		"					5.5 V	2.7 V	"	"	$\overline{G}_2$		350	"
	I_{IH9}	"	37	"	"	"	"	"	"		"					GND	GND	2.7 V	"	$\overline{P}_2$		150	"
	I_{IH10}	"	38	5.5 V	"	"	"	"	"		"					5.5 V	"	GND	"	$\overline{G}_1$		1	mA
	I_{IH11}	"	39	GND	5.5 V	"	"	"	"		"					"	"	"	"	$\overline{P}_1$		"	"
	I_{IH12}	"	40	"	GND	5.5 V	"	"	"		"					"	"	"	"	$\overline{G}_0$		"	"
	I_{IH13}	"	41	"	"	GND	5.5 V	"	"		"					GND	"	"	"	$\overline{P}_0$		"	"
	I_{IH14}	"	42	"	"	"	GND	5.5 V	"		"					"	"	"	"	$\overline{G}_3$		"	"
	I_{IH15}	"	43	"	"	"	"	GND	5.5 V		"					"	"	"	"	$\overline{P}_3$		"	"
	I_{IH16}	"	44	"	"	"	"	"	GND		"					5.5 V	"	"	"	C_n		"	"
	I_{IH17}	"	45	"	"	"	"	"	"		"					5.5 V	5.5 V	"	"	$\overline{G}_2$		"	"
	I_{IH18}	"	46	"	"	"	"	"	"		"					GND	GND	5.5 V	"	$\overline{P}_2$		"	"
	I_{OS}	3011	47		5.5 V		5.5 V				"					5.5 V		5.5 V	"	$\overline{P}$	-40	-100	"
		"	48	GND	GND	GND	GND		5.5 V	GND	"	GND					GND	GND	"	C_{n+Z}	"	"	"
		"	49	5.5 V	5.5 V	5.5 V		5.5 V			"		GND				5.5 V	5.5 V	"	$\overline{G}$	"	"	"
		"	50	GND	GND	GND	GND				"			GND		5.5 V			"	C_{n+Y}	"	"	"
		"	51				"				"				GND	5.5 V			"	C_{n+Z}	"	"	"
	I_{CCL}	3005	52	5.5 V	"	5.5 V	"	GND	GND		"					GND	5.5 V	GND	"	V_{CC}		99	"
	I_{CCH}	3005	53	GND	"	5.5 V	"	5.5 V	5.5 V		"					GND	GND	GND	"	V_{CC}		65	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{G}_1$	$\overline{P}_1$	$\overline{G}_0$	$\overline{P}_0$	$\overline{G}_3$	$\overline{P}_3$	$\overline{P}$	GND	C_{n+Z}	$\overline{G}$	C_{n+Y}	C_{n+X}	C_n	$\overline{G}_2$	$\overline{P}_2$	V_{CC}				
1 T_C = +25°C	I_{CEX}		54		GND		5.5 V		GND	5.5 V	GND							GND	5.5 V	$\overline{P}$		250	µA
			55	5.5 V	5.5 V	5.5 V	5.5 V				"	5.5 V				GND		5.5 V	"	C_{n+Z}		"	"
			56	GND	GND	GND		5.5 V	5.5 V		"		5.5 V				GND	GND	"	$\overline{G}$		"	"
			57	GND	5.5 V	5.5 V	5.5 V				"			5.5 V		GND			"	C_{n+Y}		"	"
			58			GND	5.5 V				"				5.5 V	GND			"	C_{n+X}		"	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted. V_{IL} = 0.7 V dc and $V_{OL(max)}$ = 0.45 V dc.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Functional tests 1/ 4/		59	B 2/	B 2/	B 2/	B 2/	B 2/	B 2/	L 3/	GND	H 3/	L 3/	H 3/	H 3/	B 2/	B 2/	B 2/	4.5 V	All output	H or L as shown 3/		
			60	B	B	A	A	A	"	H	"	H	L	H	L	A	A	"	"	"			
			61	A	A	B	"	A	"	"	"	L	H	L	H	B	"	"	"	"			
			62	A	B	A	"	B	A	"	"	H	L	H	H	A	"	"	"	"			
			63	B	A	"	"	A	B	"	"	H	L	H	L	B	"	"	"	"			
			64	A	B	"	"	"	A	"	"	H	H	L	L	B	B	"	"	"			
			65	B	"	B	A	"	B	"	"	L	H	H	H	A	A	A	"	"			
			66	A	"	B	"	"	"	"	"	H	L	H	H	B	B	A	"	"			
			67	"	"	A	"	"	"	"	"	L	H	L	L	A	A	B	"	"			
			68	"	"	B	"	"	"	"	"	H	L	H	H	B	A	B	"	"			
			69	"	"	A	B	"	"	"	"	H	L	"	L	B	B	A	"	"			
			70	"	A	B	B	B	"	"	"	L	"	"	H	A	A	"	"	"			
			71	"	A	A	A	A	"	"	"	H	"	"	L	B	B	"	"	"			
			72	"	B	A	B	A	"	L	"	L	H	"	L	B	A	B	"	"			
8 1/ 4/	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = +25°C	t_{PLH1}	3003 (Fig 5)	73	2.7 V	GND	2.7 V	GND	2.7 V	2.7 V		GND				OUT	IN	2.7 V	GND	5.0 V	C_n to C_{n+X}	2	12	ns
	t_{PHL1}	"	74	"	"	"	"	"	"		"				OUT	"	"	"	"	C_n to C_{n+X}	"	12.5	"
	t_{PLH1}	"	75	"	"	"	"	"	"		"			OUT		"	"	"	"	C_n to C_{n+Y}	"	12	"
	t_{PHL1}	"	76	"	"	"	"	"	"		"			OUT		"	"	"	"	C_n to C_{n+Y}	"	12.5	"
	t_{PLH1}	"	77	"	"	"	"	"	"		"	OUT				"	"	"	"	C_n to C_{n+Z}	"	12	"
	t_{PHL1}	"	78	"	"	"	"	"	"		"	OUT				"	"	"	"	C_n to C_{n+Z}	"	12.5	"
	t_{PLH2}	"	79	"	"	"	IN	"	"		"				OUT	2.7 V	"	"	"	$\overline{P}_0$ to C_{n+X}	"	9	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test no.	1 $\overline{G}_1$	2 $\overline{P}_1$	3 $\overline{G}_0$	4 $\overline{P}_0$	5 $\overline{G}_1$	6 $\overline{P}_3$	7 $\overline{P}$	8 GND	9 C_{n+Z}	10 $\overline{G}$	11 C_{n+Y}	12 C_{n+X}	13 C_n	14 $\overline{G}_2$	15 $\overline{P}_2$	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PHL2}	3003 (Fig 5)	80	2.7 V	GND	2.7 V	IN	2.7 V	2.7 V		GND				OUT	2.7 V	2.7 V	GND	5.0 V	$\overline{P}_0$ to C_{n+X}	2	9	ns
	t_{PLH2}	"	81	"	"	"	"	"	"		"			OUT		"	"	"	"	$\overline{P}_0$ to C_{n+Y}	"	"	"
	t_{PHL2}	"	82	"	"	"	"	"	"		"			OUT		"	"	"	"	$\overline{P}_0$ to C_{n+Y}	"	"	"
	t_{PLH2}	"	83	"	"	"	"	"	"		"	OUT				"	"	"	"	$\overline{P}_0$ to C_{n+Z}	"	"	"
	t_{PHL2}	"	84	"	"	"	"	"	"		"	OUT				"	"	"	"	$\overline{P}_0$ to C_{n+Z}	"	"	"
	t_{PLH2}	"	85	"	IN	"	GND	"	"		"			OUT		"	"	"	"	$\overline{P}_1$ to C_{n+Y}	"	"	"
	t_{PHL2}	"	86	"	"	"	"	"	"		"			OUT		"	"	"	"	$\overline{P}_1$ to C_{n+Y}	"	"	"
	t_{PLH2}	"	87	"	"	"	"	"	"		"	OUT				"	"	"	"	$\overline{P}_1$ to C_{n+Z}	"	"	"
	t_{PHL2}	"	88	"	"	"	"	"	"		"	"				"	"	"	"	$\overline{P}_1$ to C_{n+Z}	"	"	"
	t_{PLH2}	"	89	"	GND	"	"	"	"		"	"				"	"	IN	"	$\overline{P}_2$ to C_{n+Z}	"	"	"
	t_{PHL2}	"	90	"	"	"	"	"	"		"	"				"	"	IN	"	$\overline{P}_2$ to C_{n+Z}	"	"	"
	t_{PLH3}	"	91	"	"	IN	"	"	"		"				OUT	GND	"	GND	"	$\overline{G}_0$ to C_{n+X}	"	"	"
	t_{PHL3}	"	92	"	"	"	"	"	"		"				OUT	"	"	"	"	$\overline{G}_0$ to C_{n+X}	"	"	"
	t_{PLH3}	"	93	"	"	"	"	"	"		"			OUT		"	"	"	"	$\overline{G}_0$ to C_{n+Y}	"	"	"
	t_{PHL3}	"	94	"	"	"	"	"	"		"			OUT		"	"	"	"	$\overline{G}_0$ to C_{n+Y}	"	"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{G}_1$	$\overline{P}_1$	$\overline{G}_0$	$\overline{P}_0$	$\overline{G}_3$	$\overline{P}_3$	$\overline{P}$	GND	C_{n+Z}	$\overline{G}$	C_{n+Y}	C_{n+X}	C_n	$\overline{G}_2$	$\overline{P}_2$	V_{CC}				
9 T_C = +25°C	t_{PLH3}	3003 (Fig 5)	95	2.7 V	GND	IN	GND	2.7 V	2.7 V		GND	OUT				GND	2.7 V	GND	5.0 V	$\overline{G}_0$ to C_{n+Z}	2	9	ns
	t_{PHL3}	"	96	2.7 V	"	IN	"	"	"		"	OUT				"	"	"	"	$\overline{G}_0$ to C_{n+Z}	"	"	"
	t_{PLH3}	"	97	IN	"	2.7 V	"	"	"		"			OUT		"	"	"	"	$\overline{G}_1$ to C_{n+Y}	"	"	"
	t_{PHL3}	"	98	"	"	"	"	"	"		"			OUT		"	"	"	"	$\overline{G}_1$ to C_{n+Y}	"	"	"
	t_{PLH3}	"	99	"	"	"	"	"	"		"	OUT				"	"	"	"	$\overline{G}_1$ to C_{n+Z}	"	"	"
	t_{PHL3}	"	100	"	"	"	"	"	"		"	"				"	"	"	"	$\overline{G}_1$ to C_{n+Z}	"	"	"
	t_{PLH3}	"	101	2.7 V	"	"	"	"	"		"	"				"	IN	"	"	$\overline{G}_2$ to C_{n+Z}	"	"	"
	t_{PHL3}	"	102	"	"	"	"	"	"		"	"				"	IN	"	"	$\overline{G}_2$ to C_{n+Z}	"	"	"
	t_{PLH4}	"	103	"	IN	GND	"	"	GND		"		OUT			2.7 V	2.7 V	"	"	$\overline{P}_1$ to $\overline{G}$	"	9.5	"
	t_{PHL4}	"	104	"	IN	"	"	"	"		"		"			"	"	"	"	$\overline{P}_1$ to $\overline{G}$	"	12.5	"
	t_{PLH4}	"	105	"	GND	"	"	"	"		"		"			"	"	IN	"	$\overline{P}_2$ to $\overline{G}$	"	9.5	"
	t_{PHL4}	"	106	"	"	"	"	"	"		"		"			"	"	IN	"	$\overline{P}_2$ to $\overline{G}$	"	12.5	"
	t_{PLH4}	"	107	"	"	"	"	"	IN		"		"			"	"	GND	"	$\overline{P}_3$ to $\overline{G}$	"	9.5	"
	t_{PHL4}	"	108	"	"	"	"	"	IN		"		"			"	"	"	"	$\overline{P}_3$ to $\overline{G}$	"	12.5	"
	t_{PLH5}	"	109	"	"	IN	"	"	GND		"		"			"	"	"	"	$\overline{G}_0$ to $\overline{G}$	"	9.5	"
	t_{PHL5}	"	110	"	"	IN	"	"	"		"		"			"	"	"	"	$\overline{G}_0$ to $\overline{G}$	"	12.5	"
	t_{PLH5}	"	111	IN	"	2.7 V	"	"	"		"		"			"	"	"	"	$\overline{G}_1$ to $\overline{G}$	"	9.5	"
	t_{PHL5}	"	112	IN	"	2.7 V	"	"	"		"		"			"	"	"	"	$\overline{G}_1$ to $\overline{G}$	"	12.5	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{G}_1$	$\overline{P}_1$	$\overline{G}_0$	$\overline{P}_0$	$\overline{G}_3$	$\overline{P}_3$	$\overline{P}$	GND	C_{n+Z}	$\overline{G}$	C_{n+Y}	C_{n+X}	C_n	$\overline{G}_2$	$\overline{P}_2$	V_{CC}				
9 T_C = +25°C	t_{PLH5}	3003 (Fig 5)	113	2.7 V	GND	2.7 V	GND	2.7 V	GND		GND		OUT			2.7 V	IN	GND	5.0 V	$\overline{G}_2$ to $\overline{G}$	2	9.5	ns
	t_{PHL5}	"	114	"	"	"	"	2.7 V	"		"		"			"	IN	"	"	$\overline{G}_2$ to $\overline{G}$	"	12.5	"
	t_{PLH5}	"	115	"	"	"	"	IN	"		"		"			"	2.7 V	"	"	$\overline{G}_3$ to $\overline{G}$	"	9.5	"
	t_{PHL5}	"	116	"	"	"	"	IN	"		"		"			"	"	"	"	$\overline{G}_3$ to $\overline{G}$	"	12.5	"
	t_{PLH6}	"	117	"	"	"	IN	2.7 V	"	OUT	"					"	"	"	"	$\overline{P}_0$ to $\overline{P}$	"	8.5	"
	t_{PHL6}	"	118	"	"	"	IN	"	"	"	"					"	"	"	"	$\overline{P}_0$ to $\overline{P}$	"	12	"
	t_{PLH6}	"	119	"	IN	"	GND	"	"	"	"					"	"	"	"	$\overline{P}_1$ to $\overline{P}$	"	8.5	"
	t_{PHL6}	"	120	"	IN	"	"	"	"	"	"					"	"	"	"	$\overline{P}_1$ to $\overline{P}$	"	12	"
	t_{PLH6}	"	121	"	GND	"	"	"	"	"	"					"	"	IN	"	$\overline{P}_2$ to $\overline{P}$	"	8.5	"
	t_{PHL6}	"	122	"	"	"	"	"	"	"	"					"	"	IN	"	$\overline{P}_2$ to $\overline{P}$	"	12	"
	t_{PLH6}	"	123	"	"	"	"	"	IN	"	"					"	"	GND	"	$\overline{P}_3$ to $\overline{P}$	"	8.5	"
	t_{PHL6}	"	124	"	"	"	"	"	IN	"	"					"	"	"	"	$\overline{P}_3$ to $\overline{P}$	"	12	"
10 T_C = +125°C	t_{PLH1}	"	125	"	"	"	"	"	2.7 V		"				OUT	IN	2.7 V	"	"	C_n to C_{n+X}	"	14	"
	t_{PHL1}	"	126	"	"	"	"	"	"		"				OUT	"	"	"	"	C_n to C_{n+X}	"	14.5	"
	t_{PLH1}	"	127	"	"	"	"	"	"		"			OUT		"	"	"	"	C_n to C_{n+Y}	"	14	"
	t_{PHL1}	"	128	"	"	"	"	"	"		"			OUT		"	"	"	"	C_n to C_{n+Y}	"	14.5	"
	t_{PLH1}	"	129	"	"	"	"	"	"		"	OUT				"	"	"	"	C_n to C_{n+Z}	"	14	"
	t_{PHL1}	"	130	"	"	"	"	"	"		"	OUT				"	"	"	"	C_n to C_{n+Z}	"	14.5	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{G}_1$	$\overline{P}_1$	$\overline{G}_0$	$\overline{P}_0$	$\overline{G}_3$	$\overline{P}_3$	$\overline{P}$	GND	C_{n+Z}	$\overline{G}$	C_{n+Y}	C_{n+X}	C_n	$\overline{G}_2$	$\overline{P}_2$	V_{CC}				
10 T_C = +125°C	t_{PLH2}	3003 (Fig 5)	131	2.7 V	GND	2.7 V	IN	2.7 V	2.7 V		GND				OUT	2.7 V	2.7 V	GND	5.0 V	$\overline{P}_0$ to C_{n+X}	2	11	ns
	t_{PHL2}	"	132	"	"	"	"	"	"		"				OUT	"	"	"	"	$\overline{P}_0$ to C_{n+X}	"	"	"
	t_{PLH2}	"	133	"	"	"	"	"	"		"			OUT		"	"	"	"	$\overline{P}_0$ to C_{n+Y}	"	"	"
	t_{PHL2}	"	134	"	"	"	"	"	"		"			OUT		"	"	"	"	$\overline{P}_0$ to C_{n+Y}	"	"	"
	t_{PLH2}	"	135	"	"	"	"	"	"		"	OUT				"	"	"	"	$\overline{P}_0$ to C_{n+Z}	"	"	"
	t_{PHL2}	"	136	"	"	"	"	"	"		"	OUT				"	"	"	"	$\overline{P}_0$ to C_{n+Z}	"	"	"
	t_{PLH2}	"	137	"	IN	"	GND	"	"		"			OUT		"	"	"	"	$\overline{P}_1$ to C_{n+Y}	"	"	"
	t_{PHL2}	"	138	"	"	"	"	"	"		"			OUT		"	"	"	"	$\overline{P}_1$ to C_{n+Y}	"	"	"
	t_{PLH2}	"	139	"	"	"	"	"	"		"	OUT				"	"	"	"	$\overline{P}_1$ to C_{n+Z}	"	"	"
	t_{PHL2}	"	140	"	"	"	"	"	"		"	"				"	"	"	"	$\overline{P}_1$ to C_{n+Z}	"	"	"
	t_{PLH2}	"	141	"	GND	"	"	"	"		"	"				"	"	IN	"	$\overline{P}_2$ to C_{n+Z}	"	"	"
	t_{PHL2}	"	142	"	"	"	"	"	"		"	"				"	"	IN	"	$\overline{P}_2$ to C_{n+Z}	"	"	"
	t_{PLH3}	"	143	"	"	IN	"	"	"		"				OUT	GND	"	GND	"	$\overline{G}_0$ to C_{n+X}	"	"	"
	t_{PHL3}	"	144	"	"	IN	"	"	"		"				OUT	GND	"	GND	"	$\overline{G}_0$ to C_{n+X}	"	"	"

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases C, F Test no.	1 $\overline{G}_1$	2 $\overline{P}_1$	3 $\overline{G}_0$	4 $\overline{P}_0$	5 $\overline{G}_3$	6 $\overline{P}_3$	7 $\overline{P}$	8 GND	9 C_{n+Z}	10 $\overline{G}$	11 C_{n+Y}	12 C_{n+X}	13 C_n	14 $\overline{G}_2$	15 $\overline{P}_2$	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
10 T_C = +125°C	t_{PLH3}	3003 (Fig 5)	145	2.7 V	GND	IN	GND	2.7 V	2.7 V		GND			OUT		GND	2.7 V	GND	5.0 V	$\overline{G}_0$ to C_{n+Y}	2	11	ns
	t_{PHL3}	"	146	"	"	"	"	"	"		"			OUT		"	"	"	"	$\overline{G}_0$ to C_{n+Y}	"	"	"
	t_{PLH3}	"	147	"	"	"	"	"	"		"	OUT				"	"	"	"	$\overline{G}_0$ to C_{n+Z}	"	"	"
	t_{PHL3}	"	148	"	"	"	"	"	"		"	OUT				"	"	"	"	$\overline{G}_0$ to C_{n+Z}	"	"	"
	t_{PLH3}	"	149	IN	"	2.7 V	"	"	"		"			OUT		"	"	"	"	$\overline{G}_1$ to C_{n+Y}	"	"	"
	t_{PHL3}	"	150	"	"	"	"	"	"		"			OUT		"	"	"	"	$\overline{G}_1$ to C_{n+Y}	"	"	"
	t_{PLH3}	"	151	"	"	"	"	"	"		"	OUT				"	"	"	"	$\overline{G}_1$ to C_{n+Z}	"	"	"
	t_{PHL3}	"	152	"	"	"	"	"	"		"	"				"	"	"	"	$\overline{G}_1$ to C_{n+Z}	"	"	"
	t_{PLH3}	"	153	2.7 V	"	"	"	"	"		"	"				"	IN	"	"	$\overline{G}_2$ to C_{n+Z}	"	"	"
	t_{PHL3}	"	154	"	"	"	"	"	"		"	"				"	IN	"	"	$\overline{G}_2$ to C_{n+Z}	"	"	"
	t_{PLH4}	"	155	"	IN	"	"	"	GND		"		OUT			2.7 V	2.7 V	"	"	$\overline{P}_1$ to $\overline{G}$	"	11.5	"
	t_{PHL4}	"	156	"	IN	"	"	"	"		"		"			"	"	"	"	$\overline{P}_1$ to $\overline{G}$	"	14.5	"
	t_{PLH4}	"	157	"	GND	"	"	"	"		"		"			"	"	IN	"	$\overline{P}_2$ to $\overline{G}$	"	11.5	"
	t_{PHL4}	"	158	"	"	"	"	"	"		"		"			"	"	IN	"	$\overline{P}_2$ to $\overline{G}$	"	14.5	"
	t_{PLH4}	"	159	"	"	"	"	"	IN		"		"			"	"	GND	"	$\overline{P}_3$ to $\overline{G}$	"	11.5	"
	t_{PHL4}	"	160	"	"	"	"	"	IN		"		"			"	"	"	"	$\overline{P}_3$ to $\overline{G}$	"	14.5	"
	t_{PLH5}	"	161	"	"	IN	"	"	GND		"		"			"	"	"	"	$\overline{G}_0$ to $\overline{G}$	"	11.5	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{G}_1$	$\overline{P}_1$	$\overline{G}_0$	$\overline{P}_0$	$\overline{G}_3$	$\overline{P}_3$	$\overline{P}$	GND	C_{n+Z}	$\overline{G}$	C_{n+Y}	C_{n+X}	C_n	$\overline{G}_2$	$\overline{P}_2$	V_{CC}				
10 T_C = +125°C	t_{PHL5}	3003 (Fig 5)	162	2.7 V	GND	IN	GND	2.7 V	GND		GND		OUT			2.7 V	2.7 V	GND	5.0 V	$\overline{G}_0$ to $\overline{G}$	2	14.5	ns
	t_{PLH5}	"	163	IN	"	2.7 V	"	"	"		"		"			"	"	"	"	$\overline{G}_1$ to $\overline{G}$	"	11.5	"
	t_{PHL5}	"	164	IN	"	"	"	"	"		"		"			"	"	"	"	$\overline{G}_1$ to $\overline{G}$	"	14.5	"
	t_{PLH5}	"	165	2.7 V	"	"	"	"	"		"		"			"	IN	"	"	$\overline{G}_2$ to $\overline{G}$	"	11.5	"
	t_{PHL5}	"	166	"	"	"	"	"	"		"		"			"	IN	"	"	$\overline{G}_2$ to $\overline{G}$	"	14.5	"
	t_{PLH5}	"	167	"	"	"	"	IN	"		"		"			"	2.7 V	"	"	$\overline{G}_3$ to $\overline{G}$	"	11.5	"
	t_{PHL5}	"	168	"	"	"	"	IN	"		"		"			"	"	"	"	$\overline{G}_3$ to $\overline{G}$	"	14.5	"
	t_{PLH6}	"	169	"	"	"	IN	2.7 V	"	OUT	"					"	"	"	"	$\overline{P}_0$ to $\overline{P}$	"	10	"
	t_{PHL6}	"	170	"	"	"	IN	"	"	"	"					"	"	"	"	$\overline{P}_0$ to $\overline{P}$	"	14	"
	t_{PLH6}	"	171	"	IN	"	GND	"	"	"	"					"	"	"	"	$\overline{P}_1$ to $\overline{P}$	"	10	"
	t_{PHL6}	"	172	"	IN	"	"	"	"	"	"					"	"	"	"	$\overline{P}_1$ to $\overline{P}$	"	14	"
	t_{PLH6}	"	173	"	GND	"	"	"	"	"	"					"	"	IN	"	$\overline{P}_2$ to $\overline{P}$	"	10	"
	t_{PHL6}	"	174	"	"	"	"	"	"	"	"					"	"	IN	"	$\overline{P}_2$ to $\overline{P}$	"	14	"
	t_{PLH6}	"	175	"	"	"	"	"	IN	"	"					"	"	GND	"	$\overline{P}_3$ to $\overline{P}$	"	10	"
	t_{PHL6}	"	176	"	"	"	"	"	IN	"	"					"	"	GND	"	$\overline{P}_3$ to $\overline{P}$	"	14	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

1/ Tests shall be performed in sequence.

2/ Inputs:

A = 3.0 V minimum
B = 0.0 V or GND

3/ Outputs:

H > 1.5 V
L < 1.5 V

4/ Only a summary of attributes data is required.

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b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C, minimum.

(3) Test duration: 1,000 hours, except as permitted by Appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical tests shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

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d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - - - - Ground zero voltage potential.

V_{IN} - - - - - - - - - - - - - - - - Voltage level at an input terminal.

I_{IN} - - - - - - - - - - - - - - - - Current flowing into an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54S181
02	54S182

6.6 Manufacturers' designations. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuits	
	A	B
	Texas Instruments	Signetics Corp.
01	X	X
02	X	

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Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/78A	MICROCIRCUITS,DIGITAL,SCHOTTKY TTL,ARITHMETIC LOGIC UNIT/FUNCTION GENERATORS,MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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